

NATIONAL ENGINEERING CENTER
University of the Philippines

4.0

Agentic RAGs

Eugene Rex L. Jalao PhD

Professor

Department of Industrial Engineering and Operations Research

University of the Philippines Diliman

eljalao@up.edu.ph

UP NEC

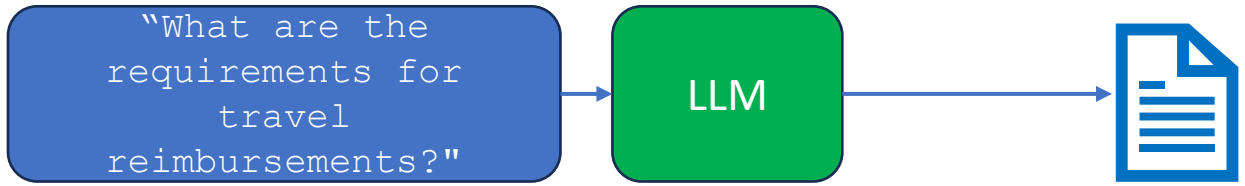
Module Outline

- Introduction to Agentic AI
  - Capstone Project Case Proposal Generation
- Deep Learning, Transformers and LLMs
  - Capstone Project Business Case Workflow
- Agentic Workflows
  - Case Study and Capstone Project Work
- **Agentic RAGs**
  - **Case Study and Capstone Project Work**
- Orchestration
  - Case Study and Capstone Project Work
- Stateful Multi-Agent Systems
  - Case Study and Capstone Project Work
- Evaluation
  - Case Study and Capstone Project Work
- LLMOps
  - Capstone Project Presentation

Outline for this Session

- Introduction to RAGs
- Preparing and Chunking Documents
- Keyword Search
- Semantic Similarity and Ranking
- Augmented Prompts
- Grounding and Evidence Sufficiency
- Citations
- Controlled RAGs
- Case Study and Project Work

Traditional LLM



Usually official receipts for plane tickets, taxi and hotel accommodations are needed for reimbursement.

- The model may not know an organization's private policies, manuals, procedures, or records.
- Its knowledge may become outdated as policies, products, regulations, or procedures change.
- It also generates answers from learned patterns rather than directly consulting a trusted source.

The Organizational Knowledge Gap

- Organizations Have Knowledge **Outside** the LLM
  - Examples of organizational knowledge include:
    - Policies and procedures.
    - Employee manuals.
    - Procurement guidelines.
    - Technical documentation.
    - Contracts and internal regulations.
    - Frequently updated operational instructions.
- The LLM may know typical reimbursement practices, but it does not automatically know the organization’s actual policy.

Three Ways to Give an LLM More Information

- Prompting
- Fine-Tuning
- RAG

Prompting Limitations

- LLMs rely on their training data and the information provided in the prompt.
- Large or frequently changing knowledge bases are impractical to include in every prompt.
- Long prompts increase token usage and may reduce focus on important information.
- Missing or outdated context can still lead to hallucinations.

Fine-Tuning

- **Definition:** Fine-tuning an LLM is the process of taking an already trained base model and training it further on a smaller, specific set of data to change its behavior or teach it a specific task
- Needs a lot of hardware resources
- New policies are not automatically added to the model after training.
- It can be difficult to trace answers to exact sources.
- Fine-tuning does not guarantee reliable recall of changing information.

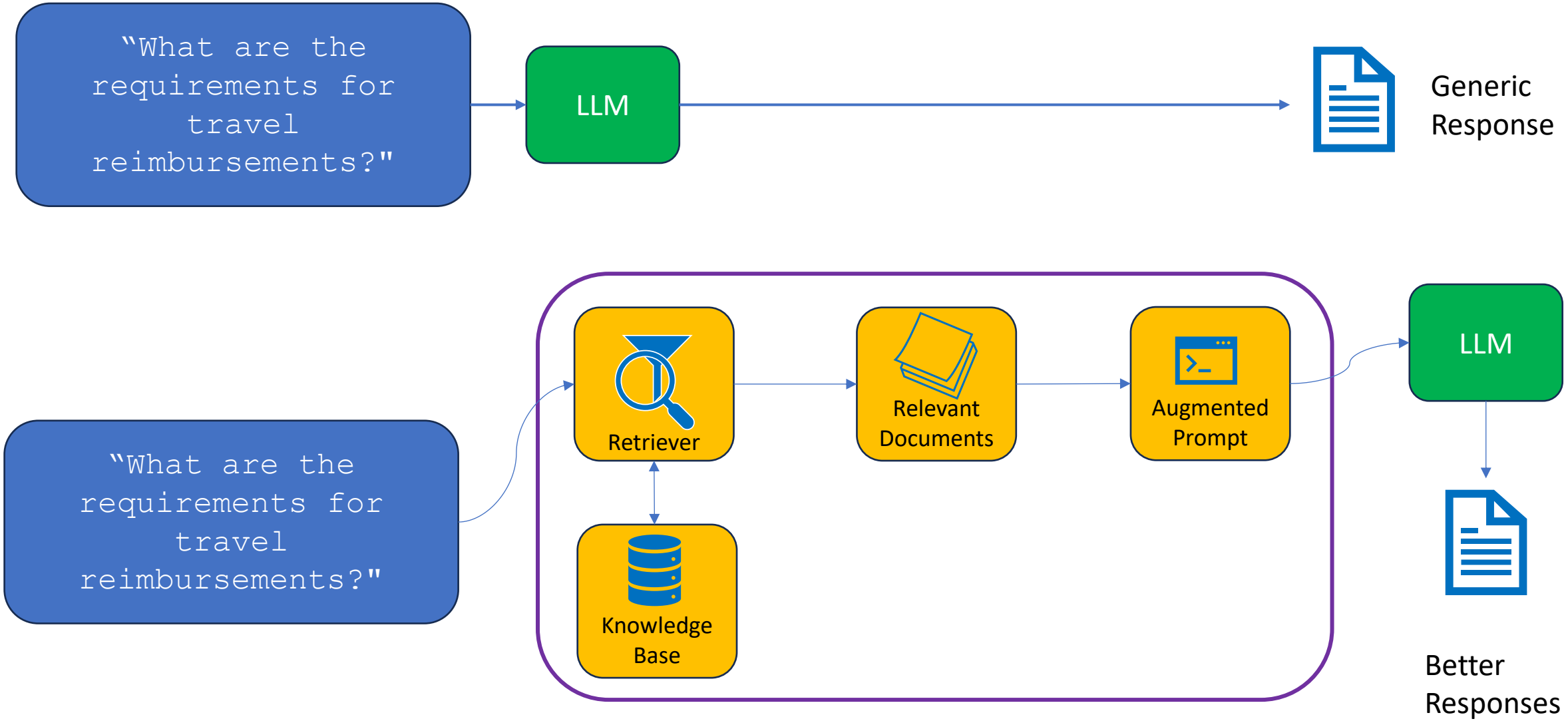
https://www.databricks.com/blog/llm-fine-tuning

Retrieval-Augmented Generation

- **Definition:** Retrieval-Augmented Generation, or RAG, is an AI framework that improves an LLM by fetching facts from external data sources before it writes an answer
  - **Ingestion:** Documents (like company files or web pages) are broken into chunks and turned into numerical codes called vector embeddings.
  - **R — Retrieval:** The system searches for relevant information.
  - **A — Augmented:** The retrieved information is added to the LLM's context.
  - **G — Generation:** The LLM generates an answer using the retrieved information.
- Instead of trying to make the LLM **memorize information**, RAG allows the LLM to **look up the relevant information when needed**

https://cloud.google.com/use-cases/retrieval-augmented-generation

RAGs and Generic LLMs



Code Generation RAG

- LLM reads your project's context
  - Functions, coding style, common modules
- Use your historical code as reference
  - RAG learns from historical code and project specific data
- Improve code generation and Q&A
  - Answers are generated based on your repository

```python
def table_adjwidths(dfdata, widths=[], align='left', size='sm', headers = True):
    # data input in dict(records)
    # widths must contain values that add up to 100

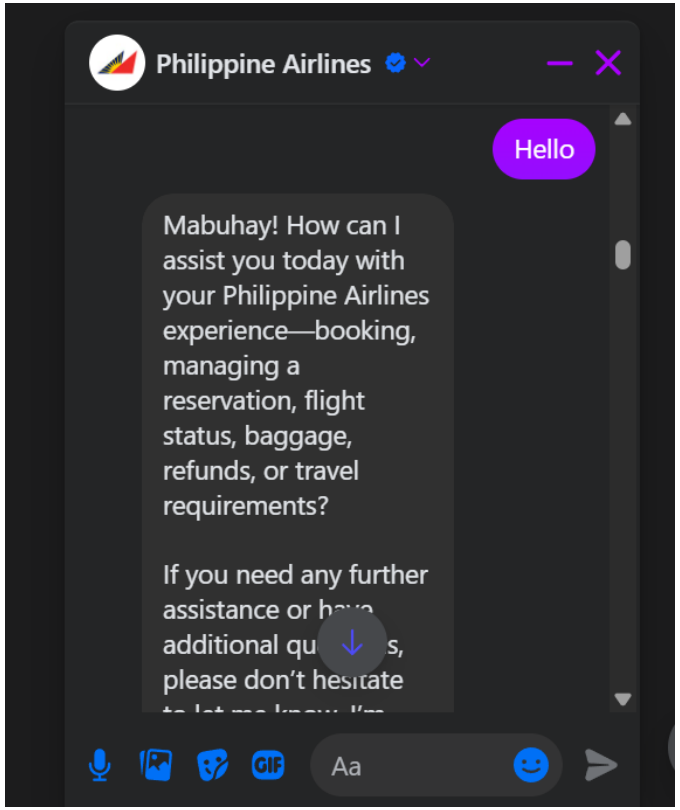
    if not isinstance(dfdata, list):
        dfdata = dfdata.to_dict('records')

    if not widths:
        widths = [100/len(dfdata[0]) for i in range(len(dfdata[0]))]
    else:
        # if the values don't add up to 100, you re-scale so the widths equal
        widths = [w/sum(widths)*100 for w in widths]

    return dbc.Table(
        [
            html.Thead(html.Tr([html.Th(colname) for colname, value in dfdata[0].items()]), className='hiddenelement' if not headers else ''),
            html.Colgroup([
                html.Col(style={'width': f"{w}%"}) for w in widths
            ]),
            html.Tbody(
                [
                    html.Tr([
                        html.Td(val) for k, val in item.items()
                    ]) for item in dfdata
                ]
            )
        ], striped=True, bordered=True, hover=True, size=size,
        style={'textAlign': align}
    )
```

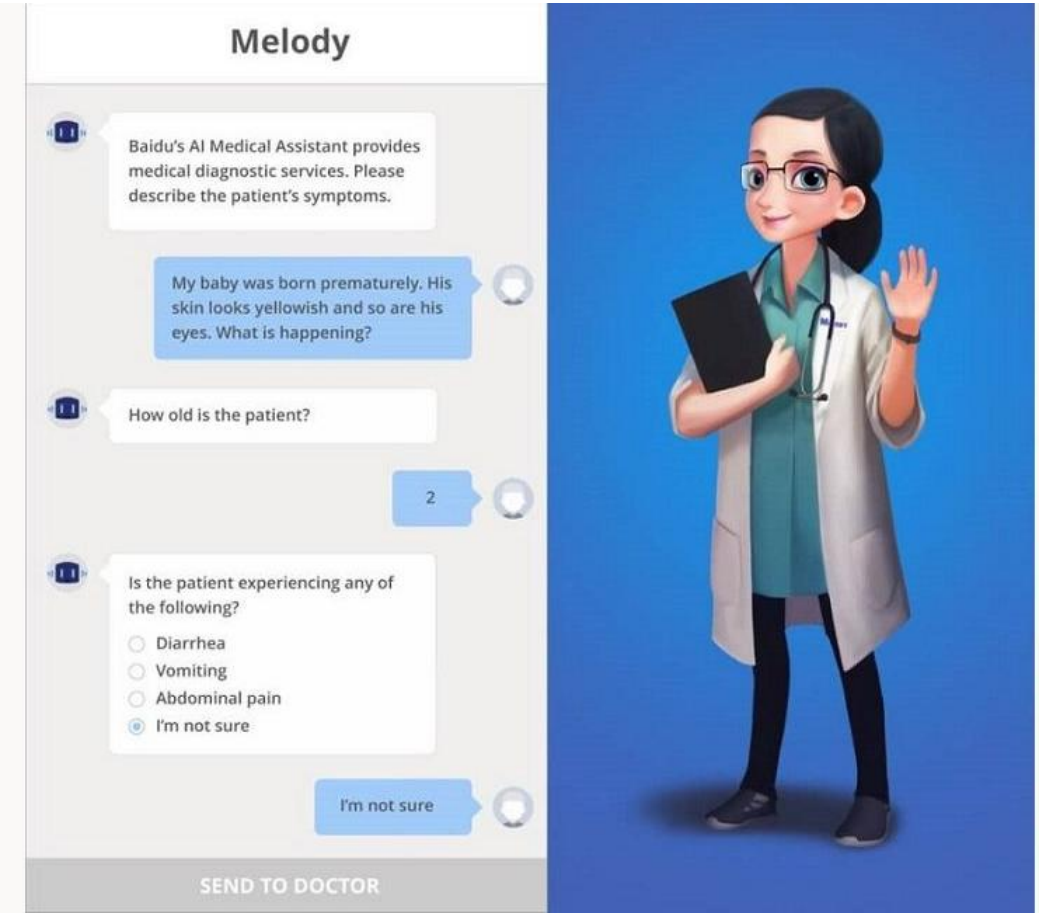
Company Chatbots

- Developed based on company requirements
  - Every company has its own policies and communication style
- Internal Documents as references
  - Company Handbook, FAQs, Historical Chats
- Grounds Answers in Real Context
  - Reduces Hallucinations and incorrect answers



Specialized Knowledge

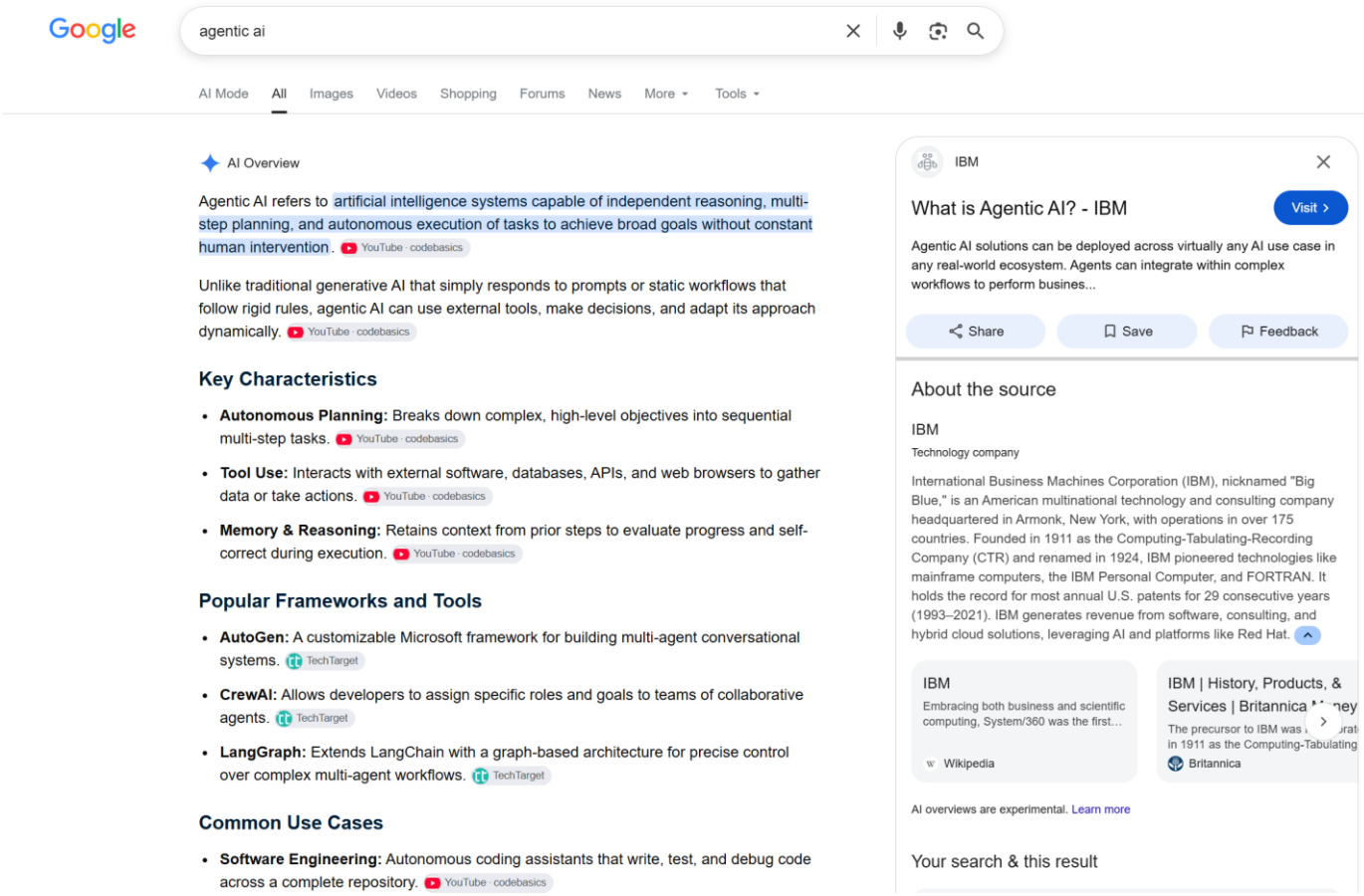
- Organizations with high impact value
  - Medical and Research Domains
- Specialized Documents as Data
  - Journal Articles, Confidential Data
- Enables Accurate and Secure Use
  - Supports privacy needs



https://govinsider.asia/intl-en/article/a-chatbot-is-helping-doctors-in-china-treat-patients

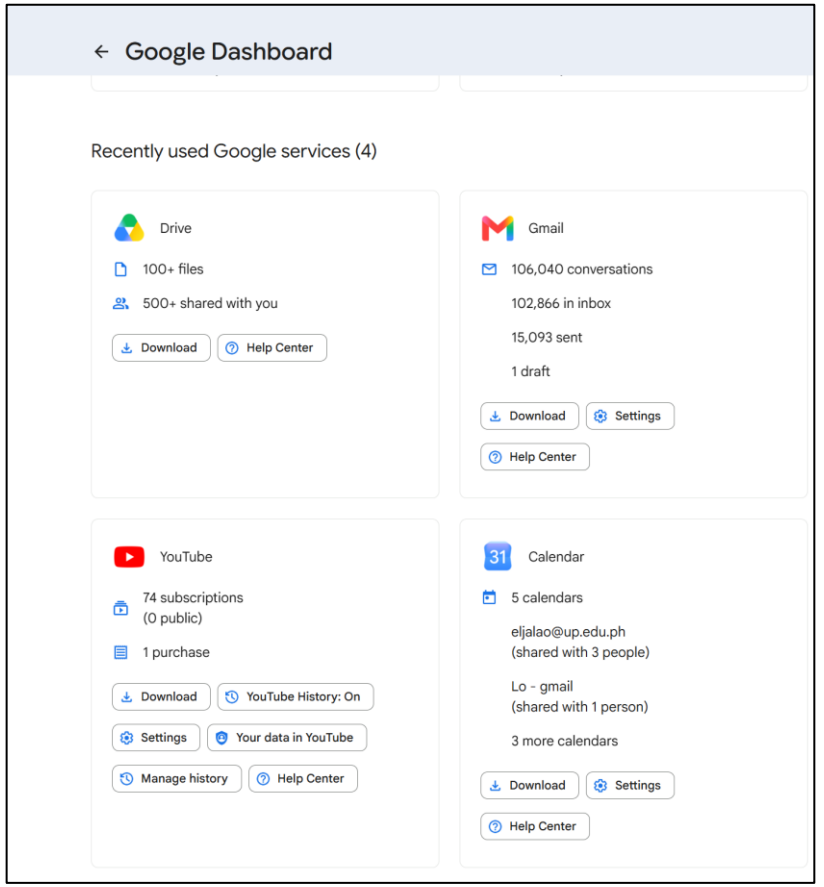
RAG Via Search Engines

- Search engines are used as retrievers
  - Return websites
- AI Summarizes Search Results
  - Summarize the websites
- Real Time Internet Data
  - Summarized are developed using latest data



Personalized RAG

- Allow the RAG to connect to your personal accounts
  - Calendar, Email, Messenger
- Let RAG summarize your day
  - Create a daily summary



LangChain

- **Definition:** **LangChain** is a Python framework for building applications that combine **LLMs with prompts, external knowledge, retrieval systems, tools, and application logic**.
- LangChain provides reusable components for connecting an LLM to other parts of an AI application.
- It supports different LLM providers, including local models running through **Ollama**.
- Components can be connected into reusable pipelines or **chains**.



https://www.langchain.com/

Introduction to RAGs

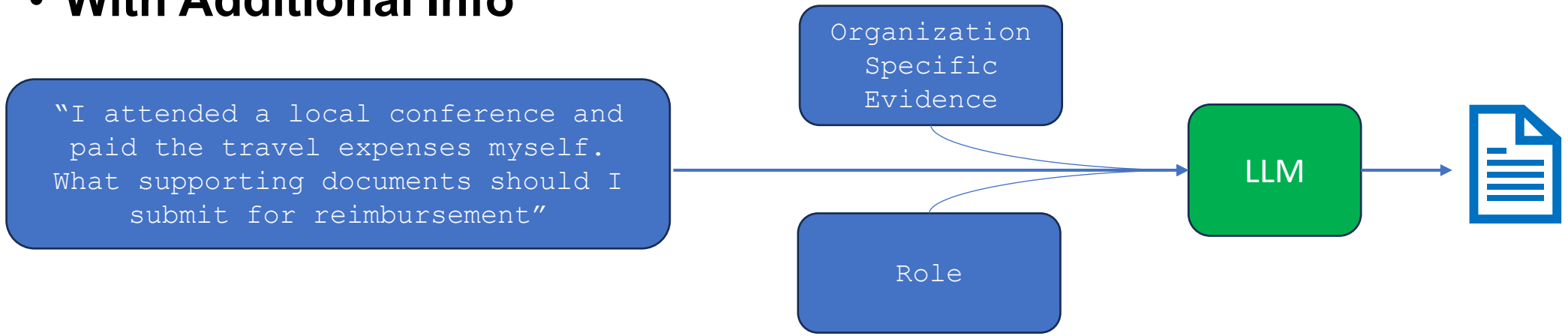
- **Exercise 1 - Introduction to RAGs**
- **Objectives**
  - Compare an LLM-only response with a RAG-grounded response.
  - Observe how organization-specific evidence changes the answer.
  - Understand the role of retrieved context in reducing unsupported responses.
  - Experience real-time **generation using Gemma**.
  - Recognize that the LLM and the organization's knowledge base are separate components.

Introduction to RAGs

- **Exercise 1 - Introduction to RAGs**
- **Business Case:**
- **LLM Only**



- **With Additional Info**



Introduction to RAGs

- **Exercise 1 - Introduction to RAGs**

```python
#%% EXERCISE 1: INTRODUCTION TO RAG
# LLM Only vs LLM + Retrieved Evidence
# Business Case: Pre-Audit Supporting Document Assistant

from langchain_ollama import ChatOllama
# ------------------------------------------------------------
# 1. LOCAL LLM THROUGH LANGCHAIN
# ------------------------------------------------------------

llm = ChatOllama(
    model="gemma3:4b",
    temperature=0
)


# ------------------------------------------------------------
# 2. USER QUESTION
# ------------------------------------------------------------

question = """
I attended a local conference and paid the travel
expenses myself.

What supporting documents should I submit
for reimbursement?
"""
```

```
USER QUESTION:

I attended a local conference and paid the travel
expenses myself.

What supporting documents should I submit
for reimbursement?


================================================================================
LLM ONLY:
--------------------------------------------------------------------------------
Okay, let's break down the supporting documents you should submit for reimbursement
of your travel expenses from the local conference, since you paid them yourself. The
specific requirements will vary slightly depending on your employer's policy, but
here's a comprehensive list, categorized for clarity:

**1. Core Documentation - Absolutely Essential:**
```

```
================================================================================
LLM + RETRIEVED EVIDENCE:
--------------------------------------------------------------------------------
To submit for reimbursement for a local conference, you should submit the following:

*   Disbursement Voucher (DV) [C01]
*   Approved Obligation Request (ObR) or Budget Utilization Request (BUR) [C01]
*   Summary of expenses [C02]
*   Invitation for conferences, seminars, or similar activities [C02]
*   Travel Order approved by the Chancellor [C02]
```

Outline for this Session

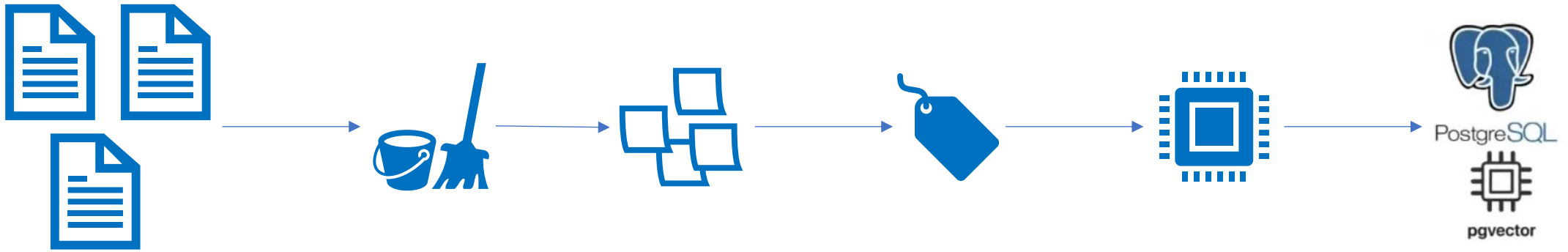
- Introduction to RAGs
- **Preparing and Chunking Documents**
- Keyword Search
- Semantic Similarity and Ranking
- Augmented Prompts
- Grounding and Evidence Sufficiency
- Citations
- Controlled RAGs
- Case Study and Project Work

Why Document Preparation Matters

- RAG systems do not search entire document collections as raw files during every question.
- Poor document preparation can cause relevant information to be missed even when the correct document exists.
- Metadata helps the system identify where retrieved information came from and whether it is still valid.
- Document preparation is therefore one of the most important steps in building a reliable RAG system
- **Definition:** Chunking is the process of breaking large documents into smaller, manageable text segments before converting them into vector embeddings for search and retrieval.

https://developer.nvidia.com/blog/finding-the-best-chunking-strategy-for-accurate-ai-responses/

Basic Document Preparation Pipeline



Documents → Cleaning → Chunking → Add Metadata → Create Embeddings → Store for Retrieval

Load and Clean the Documents

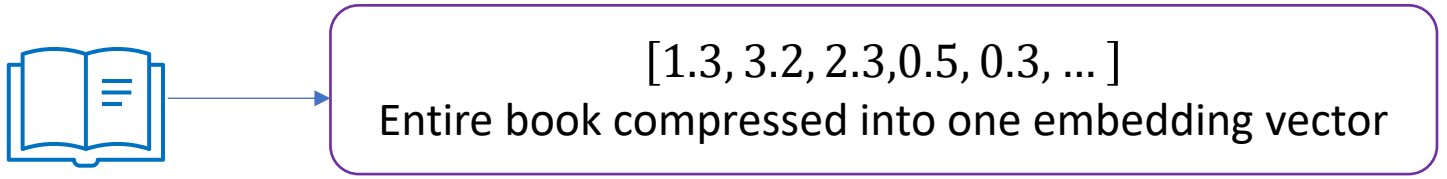
- Organizational **knowledge** may come from:
  - PDF files.
  - Word documents.
  - HTML pages.
  - Text files.
  - Spreadsheets.
  - Manuals and policy documents.
  - Frequently asked questions.
  - Internal procedures and guidelines.

Cleaning may include:

- Removing repeated page headers and footers.
- Removing unnecessary page numbers.
- Correcting broken lines or unusual spacing.
- Preserving headings and section titles.
- Removing duplicate content.
- Identifying tables or lists that require special handling.

Why Documents Must Be Chunked

- **Example**: Suppose a 50-page employee manual contains sections about, leaves, travel, benefits, procurement, and employee Conduct
  - Embedding the entire 50-page manual as one vector would make retrieval too broad.



- Issues
  - Can’t sharply represent specific topics, chapters or pages
  - Creates averaged representation of the data across all content
  - Poor search relevance since it only returns one vector
  - Retrieval returns entire book content, filling the LLM context limit

Balancing Chunk Size

- Each chunk should contain **enough information** to make sense independently.
- Each chunk receives its own embedding.
- Retrieval then ranks chunks rather than entire documents.
- A good chunk should contain one coherent idea with enough surrounding context to understand it.

Chapter Level Chunks
Too Large



- Too many topics in one vector
- Fills LLM Context Window

Word Level Chunks
Too Small



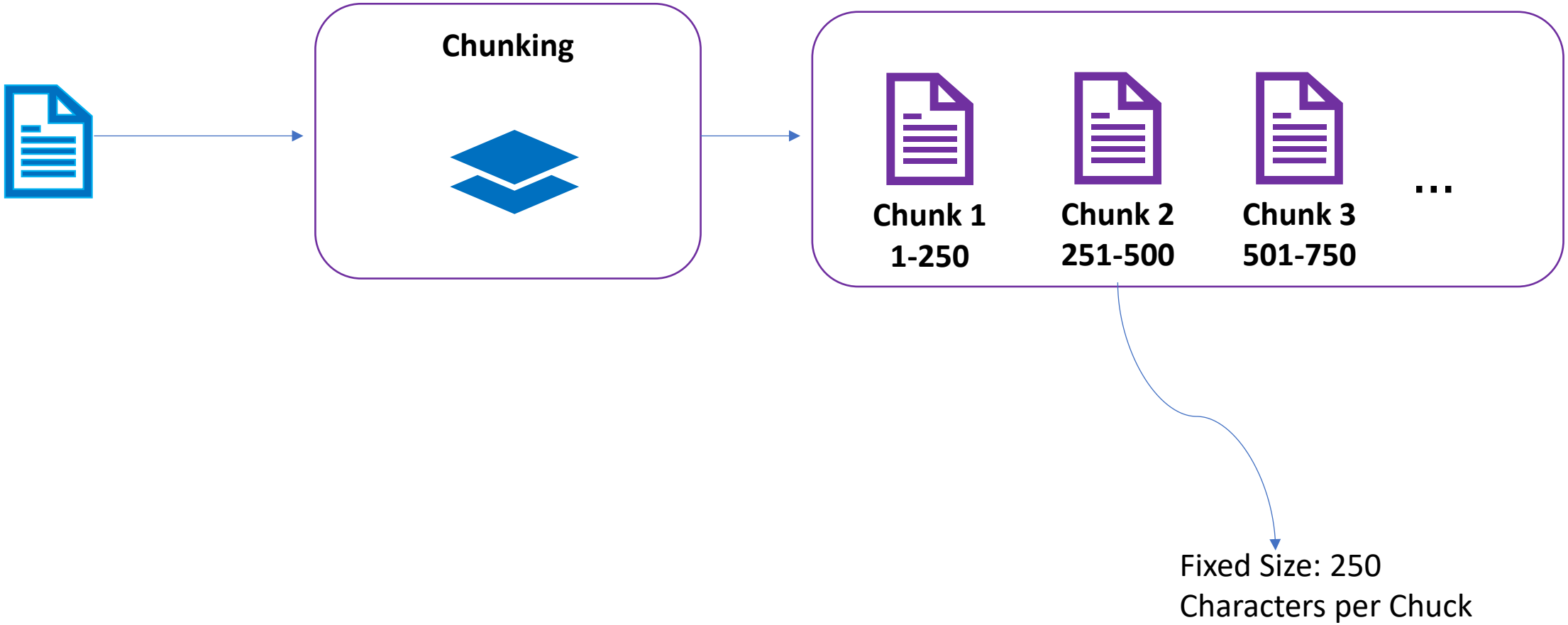
- Loses context in the sentence or paragraph
- Reduces search relevance

Just Right Size Chunks
Optimal Size



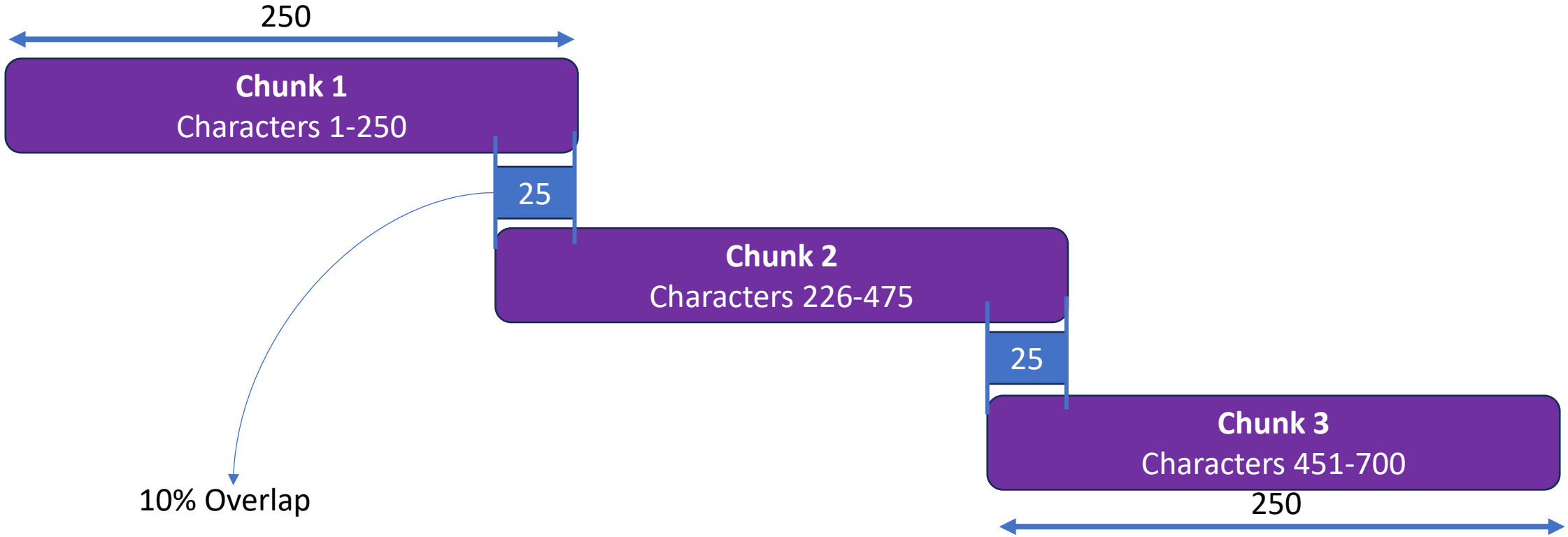
- Balance between capturing too much info or too little info per chunk

Fixed Size Chunking



Chunking with Overlap

- Minimizes words cut off from context
- Words in the middle of a chunk retain context **on each side**
- Increases relevancy but uses more chunks and space



Recursive Character Splitting

- **Splitting** text into chunks at a specified character usually “\n” new lines
- Variable chunk size but retains context of the paragraph

Admission into the University is highly competitive. At the undergraduate level, applicants compete through the UP College Admission Test (UPCAT) and/or talent test/audition. At the graduate level, aside from a Bachelor’s degree from a recognized institution of higher learning, an applicant must show proof of intellectual capacity to tackle the demands of graduate work.

A student who has earned the required number of units in another university may apply for transfer into UP. In such cases, the applicants have to satisfy the transfer admission requirements of the University as well as those of the college/degree program into which the student seeks to transfer.

No student shall be denied admission to the University System by reason of age, sex, nationality, religious belief, or political affiliation

Admission into the University is highly competitive. At the undergraduate level, applicants compete through the UP College Admission Test (UPCAT) and/or talent test/audition. At the graduate level, aside from a Bachelor’s degree from a recognized institution of higher learning, an applicant must show proof of intellectual capacity to tackle the demands of graduate work.

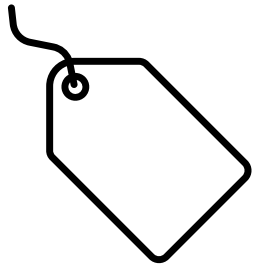
A student who has earned the required number of units in another university may apply for transfer into UP. In such cases, the applicants have to satisfy the transfer admission requirements of the University as well as those of the college/degree program into which the student seeks to transfer.

No student shall be denied admission to the University System by reason of age, sex, nationality, religious belief, or political affiliation

Add Metadata to Every Chunk

- Metadata Gives Retrieved Text Context
- Metadata can help the RAG system:
  - Identify the source document.
  - Generate citations.
  - Prefer current policies over obsolete versions.
  - Trace where an answer came from.
  - Support audit and governance requirements.
- Text provides the knowledge; metadata provides the context around that knowledge.

Admission into the University is highly competitive. At the undergraduate level, applicants compete through the UP College Admission Test (UPCAT) and/or talent test/audition. At the graduate level, aside from a Bachelor's degree from a recognized institution of higher learning, an applicant must show proof of intellectual capacity to tackle the demands of graduate work.



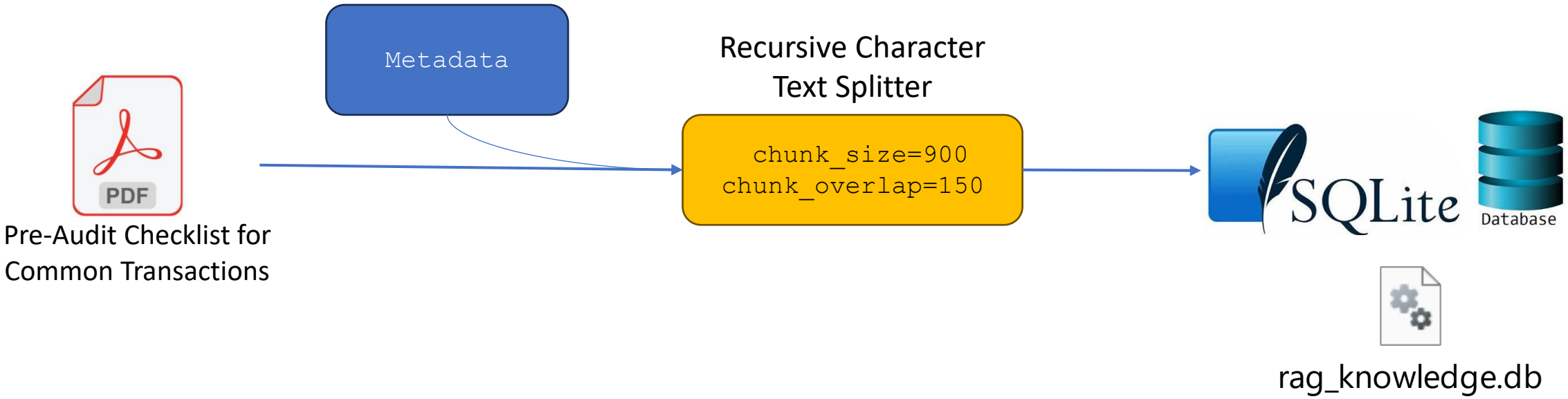
Document: UPD Acad Catalogue.pdf
Page: 10
Section: Academic Requirements
Version: 2014
URL: https://our.upd.edu.ph/acadcat.php

Preparing and Chunking Documents

- **Exercise 2: Preparing and Chunking a PDF for RAG**
- **Objectives**
  - Load a real PDF reference document using LangChain's PyPDFLoader.
  - Extract the document as individual pages while preserving page metadata.
  - Split the PDF text into smaller retrieval-ready chunks using RecursiveCharacterTextSplitter.
  - Apply chunk overlap so important context is less likely to be lost across chunk boundaries.
  - Assign a unique chunk ID and retain the original page number for later retrieval, citation, and traceability.
  - Store the chunks and metadata persistently in a **SQLite database.**
  - Inspect the resulting chunks before they are indexed for RAG.

Preparing and Chunking Documents

- **Exercise 2: Preparing and Chunking a PDF for RAG**
- **Business Case**
  - The first step in developing a **Pre-Audit RAG Assistant** is therefore to transform the PDF into smaller, traceable pieces of information that can later be searched by BM25 or SBERT and supplied to an LLM.



Exercise 2: Preparing and Chunking Documents

```python
#%% EXERCISE 2: PREPARING AND STORING A PDF FOR RAG
# PDF → Page-Aware Chunking → SQLite Knowledge Store

import os
import sqlite3
import pandas as pd

from langchain_community.document_loaders import PyPDFLoader
from langchain_text_splitters import RecursiveCharacterTextSplitter


# ------------------------------------------------------------
# 1. FILES IN THE WORKING DIRECTORY
# ------------------------------------------------------------

pdf_file = "PreAudit Checklist for Common Transactions.pdf"

database_file = "rag_knowledge.db"


# ------------------------------------------------------------
# 2. VERIFY REQUIRED FILES
# ------------------------------------------------------------

if not os.path.exists(pdf_file):
    raise FileNotFoundError(
        f"PDF file not found: {pdf_file}"
    )

if not os.path.exists(database_file):
    raise FileNotFoundError(
        f"SQLite database not found: {database_file}"
    )
```

```
================================================================================
FIRST THREE CHUNKS RETRIEVED FROM SQLITE

--------------------------------------------------------------------------------
CHUNK ID: C001
PAGE: 1
SOURCE: PreAudit Checklist for Common Transactions.pdf

TEXT:
University of The Philippines
Diliman, Quezon City
Accounting Office – Diliman
CHECKLIST OF SUPPORTING DOCUMENTS AND REQUIREMENTS
MAINTENANCE AND OTHER OPERATING EXPENSES
(MOOE)



1. Cash Advance for Local / Foreign Travel
·         Disbursement Voucher (DV)
·         Approved Obligation Request (ObR)/Budget Utilization Request (BUR)
·         Purchase request
·         Travel Order approved by the Chancellor
·         Approval of funding/request to travel by HRDO and Chancellor (as applicable)
·         Invitation (Conferences, seminars, etc.)
·         Promissory Note (settlement should be made 30 days from return date for local
travels, 60 days from return date for foreign travels)
“On or before _____, I hereby promise to settle the
amount of P_____ in full. For failure to do so, I hereby authorize

--------------------------------------------------------------------------------
CHUNK ID: C002
PAGE: 1
SOURCE: PreAudit Checklist for Common Transactions.pdf

TEXT:
“On or before _____, I hereby promise to settle the
amount of P_____ in full. For failure to do so, I hereby authorize
the University to deduct from my salary or any money due me.
I also acknowledge to have read COA Circular 96-004 dated April 19, 1996
of the Commission on Audit”
```

Outline for this Session

- Introduction to RAGs
- Preparing and Chunking Documents
- **Keyword Search**
- Semantic Similarity and Ranking
- Augmented Prompts
- Grounding and Evidence Sufficiency
- Citations
- Controlled RAGs
- Case Study and Project Work

What Is a Vector Database?

- **Definition:** A Vector Database **Stores** and Searches **Embeddings**
  - A **vector database** stores numerical embeddings together with the text and metadata they represent.
  - Each document chunk is converted into an embedding before it is stored.
  - The vector database is optimized to compare a query embedding against many stored embeddings efficiently.
  - It returns the chunks whose embeddings are most similar to the query.
  - The database does not generate answers; it only retrieves likely relevant evidence.

Pinecone







What Is Stored in a Vector Database?

- A typical stored record may contain:
  - The **embedding** is used for similarity search.
  - The **text** is the actual evidence later supplied to the LLM.
  - The **source** supports citations and traceability.
  - The **section** helps preserve document context.
  - Additional metadata can support filtering, governance, and access control.
  - The vector is useful for retrieval, but the human-readable text remains essential for answering the question.

```
{
 "id": "travel_001",
 "text": "Travel claims must be
     submitted within 10 working
     days.",
 "embedding": [0.021, -0.137, 0.084,
     ...],
 "source": "travel_policy_2026.pdf",
 "section": "Reimbursement",
 "department": "Finance",
 "version": "3.0"
}
```

Two Search Approaches

- Keyword Search
  - Looks for documents containing the exact words found in the prompt
  - Counting words for similarity measures
- Semantic Search
  - Looks for documents with similar meaning to the prompt
  - Uses embedding model for similarity

Keyword Search

- Sample prompt:

"What **supporting documents** are required for **travel reimbursement** after attending a **conference**?"



Keyword Search

Chunk A — Travel Reimbursement
"Liquidation Report / Settlement Voucher / **Reimbursement of Local/Foreign Travel** … Certificate of Travel Completed … Certificate of Appearance … Narrative Report for seminar, workshop, **conference** …"

Chunk B — Trainings / Conferences
"DV … Approved ObR/BUR … Invoice / Statement of Account … Authority / Approval to attend **training/conference** … Certificate of Participation/Attendance … Narrative Report."

Chunk C — Cash Advance for Travel
"Cash Advance for Local / Foreign **Travel** … Travel Order … Invitation for conferences and seminars … Proposed Itinerary of Travel."

Chunk D — Food Purchases
"Confirmation and Signed Attendance … Agenda / Outline of matters to be discussed … Notice of Meeting

BM25

- **Definition:** **BM25 (Best Matching 25)** is a ranking algorithm used in information retrieval systems to determine how relevant a document is to a given search query.
- It's an improved version of the traditional TF-IDF (Term Frequency–Inverse Document Frequency) approach and is widely used in modern search engines and databases.
- BM25 gives higher scores to documents that contain important query terms.
- Frequently occurring words across many documents contribute less to the score.
- Repeated occurrences of the same word increase relevance, but with diminishing benefit.
- The result is a **relevance score** used to rank documents or chunks.

https://www.geeksforgeeks.org/nlp/what-is-bm25-best-matching-25-algorithm/

How Does BM25 Compute Relevance?

- A simplified BM25 formula is:

$$BM25(D,Q) = \sum_{q \in Q} IDF(q) \frac{f(q,D)(k_1+1)}{f(q,D) + k_1\left(1 - b + b\frac{|D|}{avgdl}\right)}$$

Symbol	Meaning
q	A word or term from the query
$f(q,D)$	Number of times the term occurs in document D
$IDF(q)$	Importance of the term across the document collection
$avgdl$	Average document length
k_1	Controls the effect of repeated terms
b	Controls document-length normalization

How Does BM25 Compute Relevance?

- Prompt: "`Certificate of Appearance travel reimbursement`"

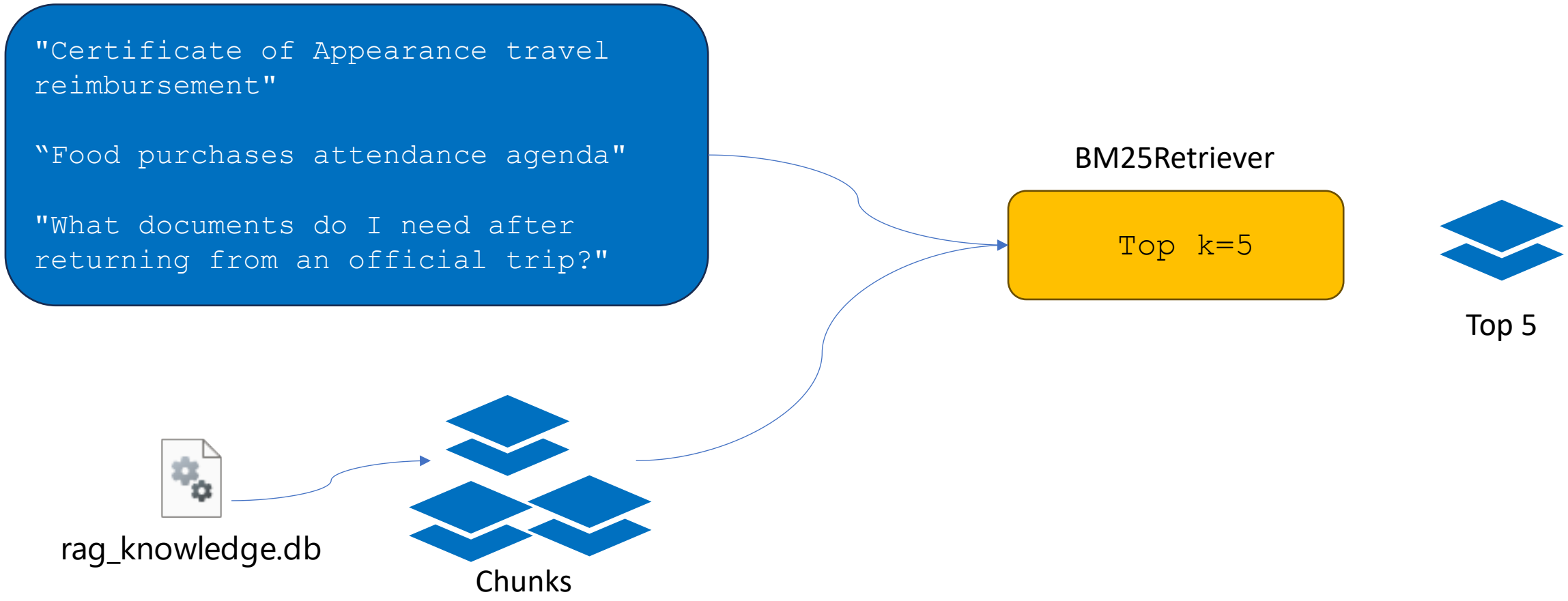
Chunk	Content	BM25 Relevance
Travel Reimbursement	"Liquidation Report / Settlement Voucher / Reimbursement of Local/Foreign Travel … Certificate of Travel Completed … Certificate of Appearance …"	Very High
Trainings / Conferences	"Certificate of Participation/Attendance … Certificate of travel completed … Itinerary of travel."	Medium
Cash Advance for Travel	"Cash Advance for Local / Foreign Travel … Travel Order … Proposed Itinerary of Travel."	Medium–Low
Food Purchases	"Confirmation and Signed Attendance … Agenda … Notice of Meeting …"	Low

Keyword Search

- **Exercise 3: Keyword Retrieval with LangChain BM25**
- **Objectives**
  - Connect to the persistent **SQLite RAG knowledge store**
  - Retrieve stored document chunks and metadata without reprocessing the PDF.
  - Convert SQLite records into LangChain Document objects.
  - Build a **BM25 keyword index** from the stored chunks.
  - Calculate BM25 relevance scores and rank the **Top-k** results.
  - Inspect which query terms matched each retrieved chunk.
  - Observe the strengths and limitations of lexical retrieval.

Keyword Search

- **Exercise 3: Keyword Retrieval with LangChain BM25**
- **Test Queries**



Keyword Search

```python
#%% EXERCISE 3: KEYWORD SEARCH WITH BM25
# SQLite Helper → LangChain Documents → BM25 Scoring → Ranked Chunks

import os
import re
import sqlite3

from langchain_core.documents import Document
from langchain_community.retrievers import BM25Retriever


# ============================================================
# HELPER FUNCTION: LOAD CHUNKS FROM SQLITE
# ============================================================

def load_chunks_from_sqlite(
    database_file="rag_knowledge.db"
):

    # ------------------------------------------------------
```

```
Chunks loaded from SQLite: 30

================================================================================
QUERY:
Certificate of Appearance travel reimbursement

QUERY TOKENS:
['certificate', 'of', 'appearance', 'travel', 'reimbursement']

BM25 RANKING:

--------------------------------------------------------------------------------
Rank 1: [C004] Page 2
BM25 Score: 7.5708
Matched Query Terms: ['appearance', 'certificate', 'of', 'reimbursement', 'travel']
Source: PreAudit Checklist for Common Transactions.pdf

TEXT:
2.                  Liquidation Report / Settlement Voucher/ Reimbursement of Local/Foreign Travel
              (Subject to Executive Order No. 77)
·         Liquidation Report – if Liquidation or Settlement
·         DV and approved ObR/BUR – if reimbursement
·         Summary of expenses
·         Invitation (Conferences, seminars, etc.)
·         Travel Order approved by the Chancellor
·         Travel authority endorsed by the Chancellor to the Department of Foreign Affairs
(DFA) – for travels abroad of University employees
·         Actual Itinerary of Travel
·         Copy of UNDP DSA rate
·         Copy of the basis for the foreign exchange rate used to convert foreign currency to
Philippine peso
·         Certificate of Travel Completed
·         Certificate of Appearance
·         Original Boarding pass, terminal fee stub

--------------------------------------------------------------------------------
Rank 2: [C005] Page 2
BM25 Score: 7.4381
Matched Query Terms: ['appearance', 'certificate', 'of', 'reimbursement', 'travel']
Source: PreAudit Checklist for Common Transactions.pdf
```

Outline for this Session

- Introduction to RAGs
- Preparing and Chunking Documents
- Keyword Search
- **Semantic Similarity and Ranking**
- Augmented Prompts
- Grounding and Evidence Sufficiency
- Citations
- Controlled RAGs
- Case Study and Project Work

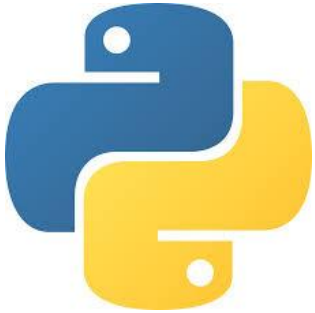
Limitations of Keyword Search

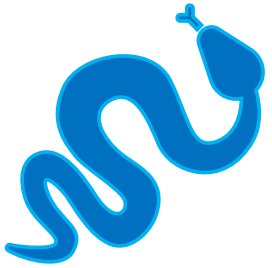
- To improve on lexical search, it's necessary to capture not only the presence or absence of a word but its **meaning**
- Cannot Match Synonyms:

Glad

Happy

- Incorrectly matches different meanings





What Is Semantic Search?

- **Definition:** A **semantic search engine** understands the meaning and intent behind a user's query rather than simply matching exact words. It does this by analyzing how words relate to one another and how they are commonly used in natural language
  - Traditional keyword search asks, **“Do these words appear in the document?”**
  - Semantic search asks, **“Does this document mean something similar to the question?”**
  - The user's query is converted into a numerical representation called an **embedding**.
  - Document chunks are also represented as embeddings.
  - The system searches for document embeddings that are mathematically close to the query embedding.
  - This allows the system to retrieve relevant passages even when different words or phrases are used

https://kb.wisc.edu/kbGuide/142365

Semantic Search

- Example:

Chunk A — Travel Reimbursement

- "Liquidation Report / Settlement Voucher / Reimbursement of Local/Foreign Travel … Actual Itinerary of Travel … Certificate of Travel Completed … Certificate of Appearance …"

Chunk B — Food Purchases

- "Liquidation Report / Settlement Voucher / Reimbursement of Local/Foreign Travel … Actual Itinerary of Travel … Certificate of Travel Completed … Certificate of Appearance …"

Chunk C — Repairs and Maintenance

- "Invoice … Inspection Report … Pre-repair Inspection … Report of Waste Material … Post Repair Inspection Report …"

User query:

"I just came back from an official trip and need to claim my expenses. What paperwork should I prepare?"

Expected ranking:

1. Chunk A← Most relevant
2. Chunk B
3. Chunk C

Retrieval Embeddings

- A retrieval embedding represents meaning as a **vector**
  - A retrieval embedding is a numerical vector that represents the semantic meaning of a piece of text.
  - Each dimension contains a learned numerical feature rather than a directly interpretable word or concept.
  - Texts with similar meanings should have nearby vector representations.
  - Texts with unrelated meanings should be farther apart.
  - The embedding model is trained specifically to make semantic comparison useful.
  - The same model should normally encode both the documents and the user queries.
- **Example:**

Text:

`"Employees receive 15 days of annual leave."`

↓ Embedding Model

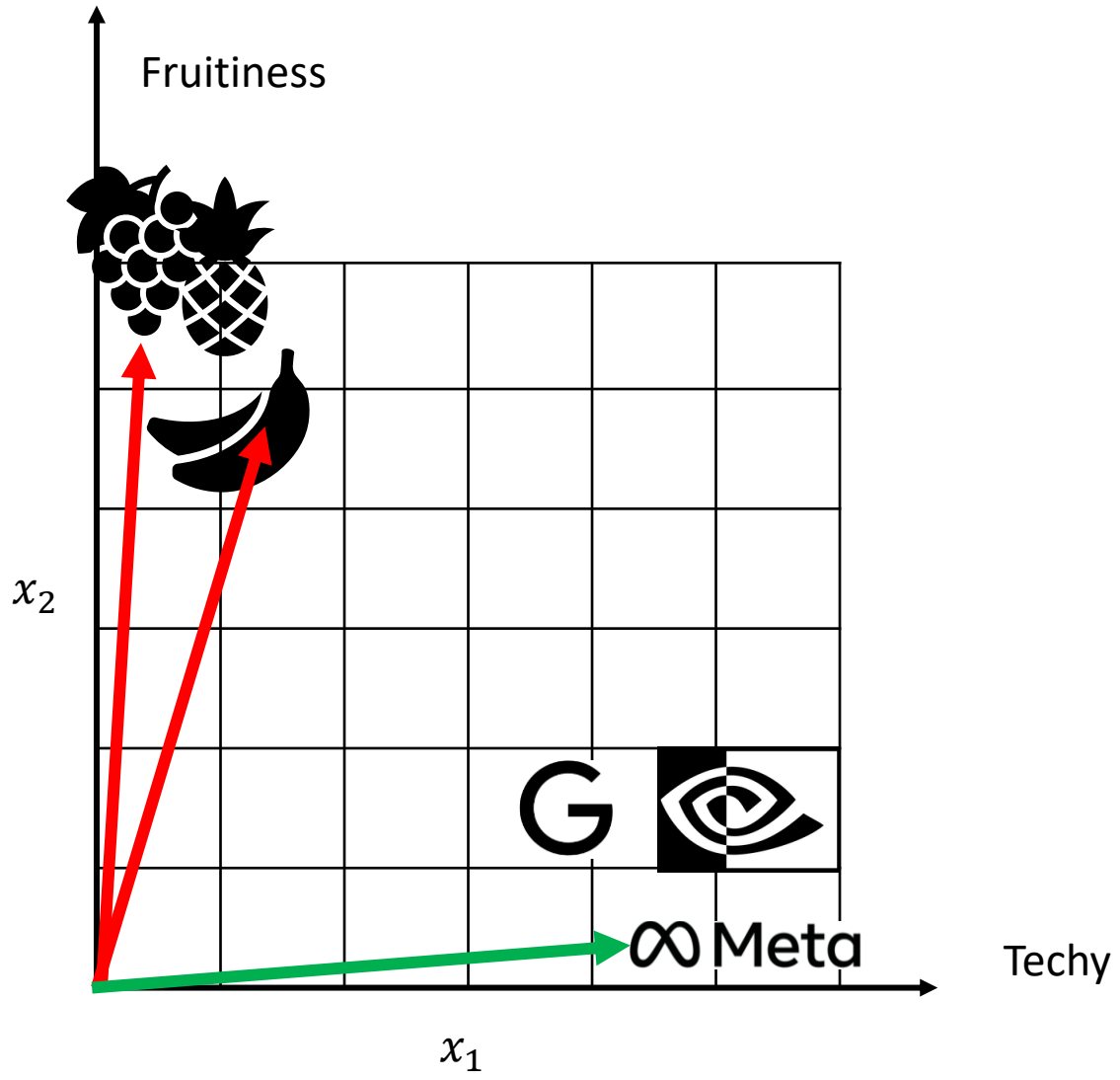
[0.021, -0.137, 0.084, ..., 0.052]

Measuring Similarity with Cosine Similarity

- A common similarity measure is **cosine similarity**.
- For two vectors $A$and B:

$$cosine\ similarity(A, B) = \frac{A \cdot B}{\| A \| \| B \|}$$

- For most modern retrieval embeddings, the practical comparison is usually between relatively high and low similarity scores



Keyword Search vs Semantic Search

- Exact Words vs Similar Meaning

Keyword Search	Semantic Search
Looks for matching words	Looks for similar meaning
Strong when exact terms are known	Strong when wording differs
Easy to understand	Better for natural-language queries
Can miss synonyms	Can match paraphrases
Useful for IDs, names, and exact phrases	Useful for policies, FAQs, and knowledge retrieval

- RAG often benefits from **semantic search** because users rarely phrase questions exactly like the source documents.

Metadata Filtering

- **Definition:** Metadata filtering uses structured information about documents—such as document type, date, department, product, customer segment, or policy version—to limit retrieval to the most relevant subset of the knowledge base.
  - **Semantic search asks: “Which chunks mean something similar?”**
  - **Metadata filtering asks: “Which chunks are even eligible to be searched?”**




```
result = articles[
(articles["section"]== "Opinion") &
(articles["author"]== "Rex Jalao") &
(articles["date"]>= "2026-01-01") &
(articles["date"]<= "2026-06-30")
 ]
```

Top-k Retrieval

- Retrieve the **most similar chunks**
  - top_k specifies how many results the search should return.
  - A small top_k may miss useful supporting information.
  - A very large top_k may introduce irrelevant context.
- More retrieved text is not always better.
  - The optimal value depends on chunk size, document structure, and the type of question.
  - top_k should therefore be evaluated rather than selected arbitrarily.
- Retrieve enough evidence to answer the question, but not so much that irrelevant context overwhelms the model.

- Query: `"I just returned from an official trip. What documents do I need to claim my expenses?"`

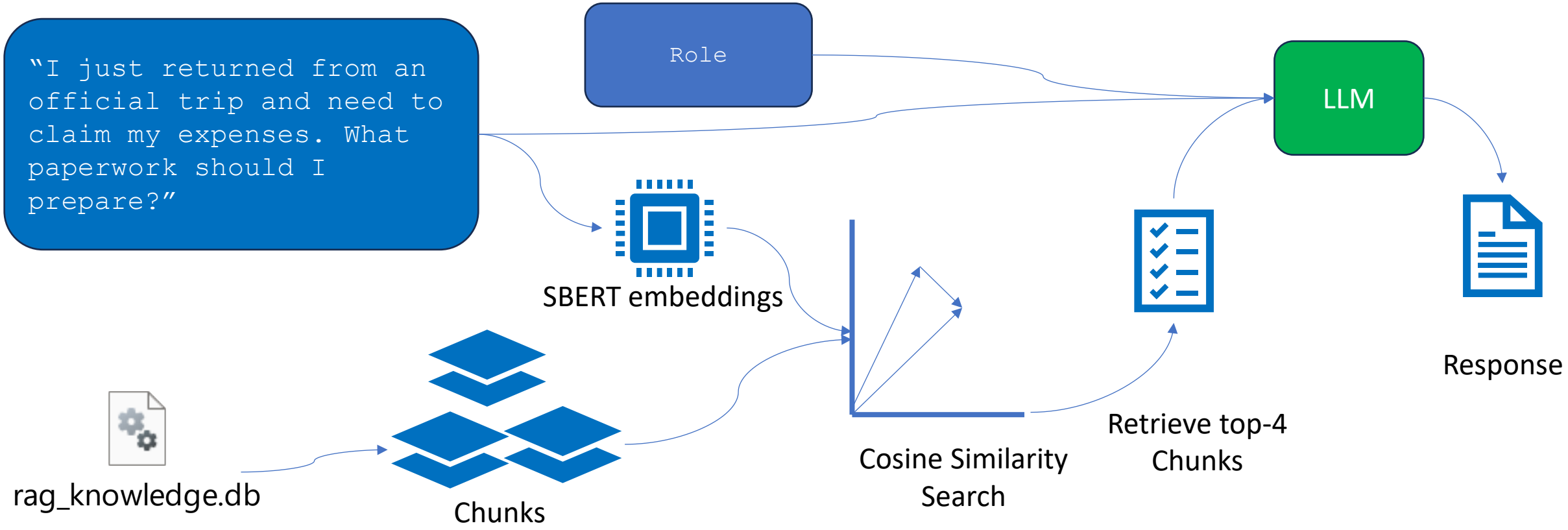
- Possible result: top_k = 3
- Rank 1: Travel Reimbursement
  - `Similarity: 0.86`
  - `"Liquidation Report / Settlement Voucher / Reimbursement of Local/Foreign Travel … Summary of expenses … …"`
- Rank 2: Trainings / Conferences
  - `Similarity: 0.71`
  - `"DV … Approved ObR/BUR … Invoice / Statement of Account … Authority / Approval to attend …`
- Rank 3:Cash Advance for Local / Foreign Travel
  - `Similarity: 0.44`
  - `"Disbursement Voucher … Travel Order … Invitation … Promissory Note …`

Semantic Similarity and Ranking

- **Exercise 4: Search by Meaning with LangChain**
- **Objectives**
  - Load previously ingested chunks from the **SQLite RAG knowledge store**.
  - Use **SBERT** to convert stored chunks and user queries into embeddings.
  - Calculate **cosine similarity** between query and document vectors.
  - Rank the most semantically relevant chunks.
  - Compare direct SBERT retrieval with a **LangChain vector store**.
  - Recognize how semantic search can retrieve relevant evidence even when users use different wording.

Semantic Similarity and Ranking

- **Exercise 4: Search by Meaning with LangChain**
- Business Case



Semantic Similarity and Ranking

```python
#%% EXERCISE 4: SEMANTIC SIMILARITY AND RANKING
# SQLite Helper → SBERT Embeddings → Semantic Search
# → LangChain Vector Retrieval

import os
import sqlite3
import pandas as pd

from sentence_transformers import SentenceTransformer

from langchain_core.documents import Document
from langchain_huggingface import HuggingFaceEmbeddings
from langchain_core.vectorstores import InMemoryVectorStore

# ------------------------------------------------------------
# 1. LOAD CHUNKS FROM SQLITE
# ------------------------------------------------------------

chunks = load_chunks_from_sqlite(
    "rag_knowledge.db"
)

print(
    "Chunks loaded from SQLite:",
    len(chunks)
)


# ------------------------------------------------------------
# 2. TEST QUERIES
# ------------------------------------------------------------
# These queries use conversational wording rather than
# exact phrases found in the checklist.
```

```
--------------------------------------------------------------------------------
Rank 3: [C016] Page 6
(Expenses worth P 300.00 to P 1,000.00)
·         Taxi Receipt
·         UBER / Grab
·         Justification on the need to avail the transport service over customary modes of
transportation
·         Canvass from 3 reputable suppliers for rentals P1,000.00 and above
·         Certification of one-time reimbursement for E-receipts (Grab/Lalamove/Mr.
Speedy/other similar categories)
·         Certification that courier expenses are for official use
·         Additional requirements for

--------------------------------------------------------------------------------
Rank 4: [C004] Page 2
2.                  Liquidation Report / Settlement Voucher/ Reimbursement of Local/Foreign Travel
            (Subject to Executive Order No. 77)
·         Liquidation Report – if Liquidation or Settlement
·         DV and approved ObR/BUR – if reimbursement
·         Summary of expenses
·         Invitation (Conferences, seminars, etc.)
·         Travel Order approved by the Chancellor
·         Travel authority endorsed by the Chancellor to the Department of Foreign Affairs
(DFA) –

================================================================================
GROUNDED GEMMA RESPONSE:
--------------------------------------------------------------------------------
To submit for reimbursement, you should provide the following:

*   Certificate of Appearance
*   Invoice of Airfare / Registration Fee
*   Narrative Report (For seminar, workshop, conference)
*   Taxi Receipt / UBER / Grab (if applicable)
*   Justification on the need to avail the transport service over customary modes of transportation (if applicable)
```

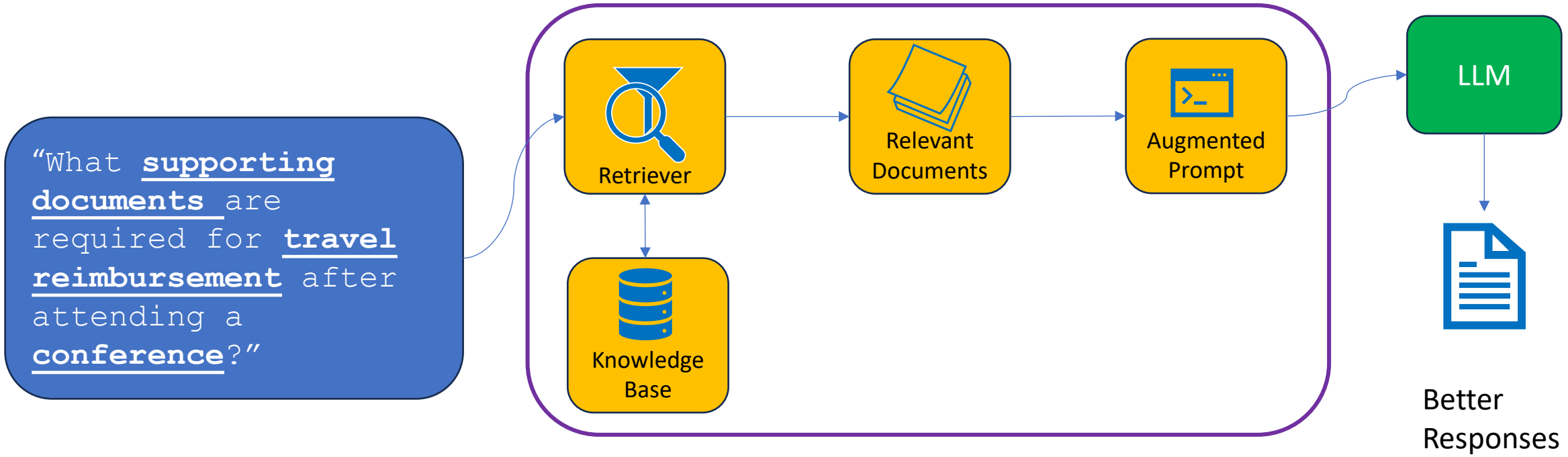
Outline for this Session

- Introduction to RAGs
- Preparing and Chunking Documents
- Keyword Search
- Semantic Similarity and Ranking
- **Augmented Prompts**
- Grounding and Evidence Sufficiency
- Citations
- Controlled RAGs
- Case Study and Project Work

Building the Augmented Prompt

- After retrieval, the RAG system has the **user question** and one or more **relevant chunks** from the knowledge base.
- These retrieved chunks are inserted into the prompt sent to the LLM.
- The resulting prompt is called an **augmented prompt** because the original question has been enriched with external information.
- The LLM is instructed to answer using the retrieved context rather than relying only on its pretrained knowledge.
- This is the **Augmented** part of Retrieval-Augmented Generation.

RAG System


"What supporting documents are required for travel reimbursement after attending a conference?"
Retriever
Knowledge Base
Relevant Documents
Augmented Prompt
LLM
Better Responses

Anatomy of an Augmented Prompt

- A typical augmented prompt contains four parts:

Augmented Prompt

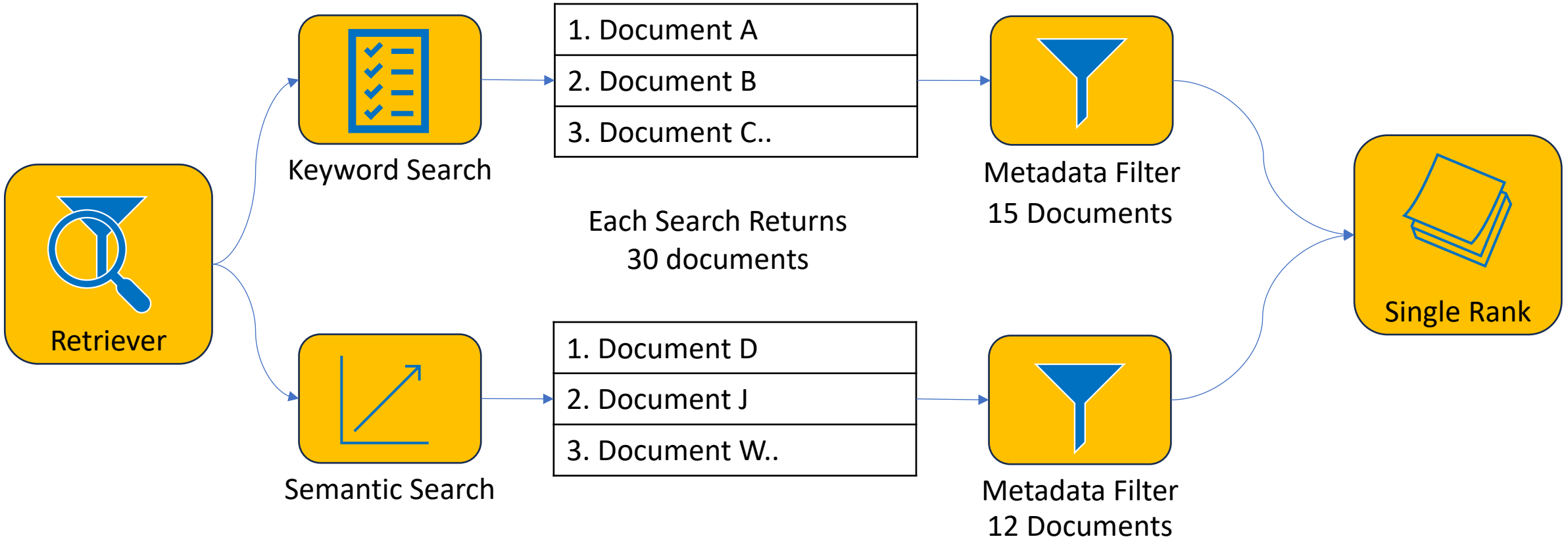
1. SYSTEM / TASK INSTRUCTIONS
"You are a pre-audit assistant for a University Accounting Office. Answer using only the retrieved checklist evidence. Do not invent documentary requirements or procedures."
2. RETRIEVED CONTEXT
Chunk A — Travel Reimbursement
DV and approved ObR/BUR, summary of expenses, Travel Order, Actual Itinerary of Travel, Certificate of Travel Completed, Certificate of Appearance, and applicable travel receipts. Pre-Audit Checklist for Common …
Chunk B — Conference Requirement
Narrative Report is required for seminar, workshop, conference, and strategic planning activities. Pre-Audit Checklist for Common …
Chunk C — Trainings / Conferences
Authority/approval to attend, Certificate of Participation/Attendance, Narrative Report, and additional outside-campus travel documents. Pre-Audit Checklist for Common …
3. USER QUESTION
"I attended a local conference and paid the travel expenses myself. What supporting documents should I submit for reimbursement?"
4. OUTPUT INSTRUCTIONS
"Give a concise checklist of the supported requirements. Cite the relevant chunk and page for each major requirement. If the retrieved evidence is incomplete, say so."

Building an Effective Augmented Prompt

- Clearly separate the **instructions**, **retrieved context**, and **user question**.
- Tell the LLM that the retrieved context is the authoritative source for the answer.
- Include only chunks that are relevant to the question.
- Preserve useful metadata such as document title, section, date, or source ID.
- Tell the LLM what to do when the retrieved information is insufficient.
- Specify the desired response format.
- Ask the model to cite or identify the retrieved sources when appropriate.
- Avoid adding excessive irrelevant context because it can distract the model.

Hybrid search

- Hybrid search is a retrieval method that combines keyword-based search, with semantic or vector search using embeddings, then merges or re-ranks the results to find the most relevant information.



Reciprocal Rank Fusion (RRF)

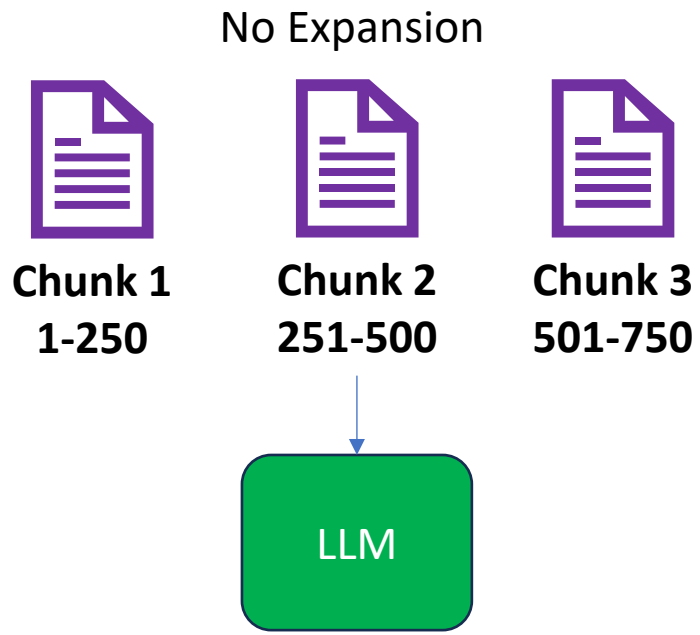
- BM25 and SBERT scores use **different numerical scales**.
- RRF combines their **rank positions** instead of averaging incompatible scores.
- Documents ranked highly by either retriever can still receive a strong final score.
- Documents ranked highly by **both** methods tend to rise to the top.

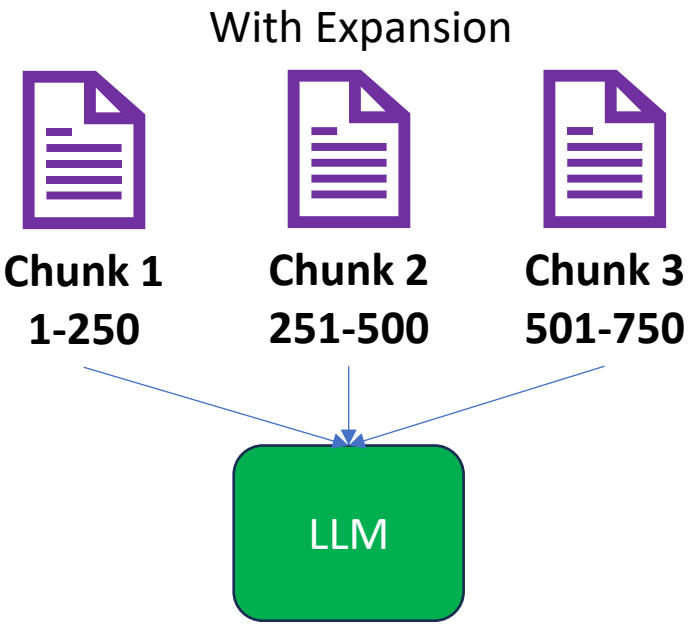
$$RRF(d) = \sum_{r} \frac{w_r}{k + rank_r(d)}$$

- d= document or chunk
- $rank_r(d)$ = rank of chunk d from retriever r
- w_r = weight assigned to that retriever
- k= smoothing constant, commonly **60**

Neighbor Expansion

- If retrieval returns only one highly ranked chunk, important requirements may remain in the chunk immediately before or after it.
- Neighbor expansion restores surrounding context that may have been split during chunking.



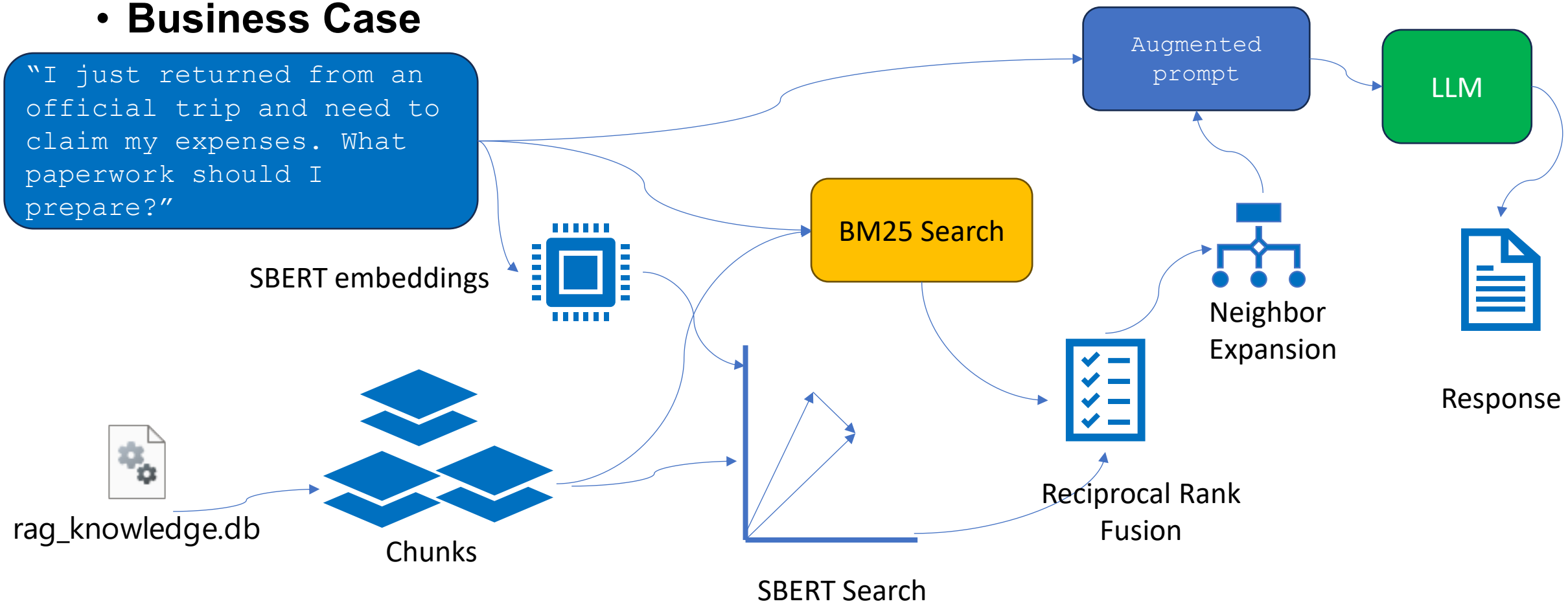


Augmented Prompts

- **Exercise 5: Hybrid Retrieval and Augmented Prompts**
- **Objectives**
  - Retrieve stored chunks from the **SQLite knowledge base**.
  - Apply **BM25** to identify strong keyword matches.
  - Apply **SBERT** to identify semantically similar chunks.
  - Combine both rankings using **Reciprocal Rank Fusion (RRF)**.
  - Select the strongest Top-k evidence from the hybrid ranking.
  - Insert retrieved evidence into a LangChain augmented prompt.
  - Generate a grounded response using **Gemma 3 4B**.

Augmented Prompts

- **Exercise 5: Hybrid Retrieval and Augmented Prompts**
- **Business Case**



Exercise 5 — Augmented Prompts

```python
#%% EXERCISE 6: GROUNDING AND EVIDENCE SUFFICIENCY
# SQLite → BM25 + SBERT → Reciprocal Rank Fusion
# → Neighbor Expansion → Gemma Evidence Evaluation
# → SUFFICIENT / PARTIAL / INSUFFICIENT
# → Compare with Expected Evaluation Label

import os
import re
import sqlite3
import numpy as np

from langchain_core.documents import Document
from langchain_core.prompts import ChatPromptTemplate
from langchain_core.output_parsers import JsonOutputParser

from langchain_community.retrievers import BM25Retriever
from langchain_huggingface import HuggingFaceEmbeddings
from langchain_ollama import ChatOllama


def hybrid_retrieve(
    question,
    chunks,
    bm25_retriever,
    embeddings,
    chunk_embeddings,
    top_k=4,
    neighbor_window=1,
    keyword_weight=0.50,
    semantic_weight=0.50,
    rrf_constant=60
):
```

```
--------------------------------------------------------------------------
Hybrid Rank 4: [C004] Page 2
BM25 Rank: 4
BM25 Score: 8.4069
Semantic Rank: 4
Semantic Similarity: 0.4081
Hybrid RRF Score: 0.015625
Matched Keywords: ['and', 'expenses', 'for', 'local', 'reimbursement', 'the', 'travel']
Source: PreAudit Checklist for Common Transactions.pdf

TEXT PREVIEW:
2.                  Liquidation Report / Settlement Voucher/ Reimbursement of Local/Foreign Travel
             (Subject to Executive Order No. 77)
·         Liquidation Report – if Liquidation or Settlement
·         DV and approved ObR/BUR – if reimbursement
·         Summary of expenses
·         Invitation (Conferences, seminars, etc.)
·         Travel Order approved by the Chancellor
·         Travel authority endorsed by the Chancellor to the Department of Foreign Affairs
(DFA) – for travels abroad of University employees
·         Actual Itinerary of Travel
·         Copy of

================================================================================
GROUNDED GEMMA RESPONSE:
--------------------------------------------------------------------------------
Based on the available evidence, you should submit the following documents:

*   Disbursement Voucher (DV)
*   Approved Obligation Request (ObR)/Budget Utilization Request (BUR)
*   Purchase request
*   Travel Order approved by the Chancellor
*   Invitation (Conferences, seminars, etc.)
*   Summary of expenses
*   Certificate of Travel Completed
```

Outline for this Session

- Introduction to RAGs
- Preparing and Chunking Documents
- Keyword Search
- Semantic Similarity and Ranking
- Augmented Prompts
- **Grounding and Evidence Sufficiency**
- Citations
- Controlled RAGs
- Case Study and Project Work

Grounding: Answer Using the Retrieved Evidence

- Grounding means that the LLM generates its answer using the **retrieved evidence** rather than relying mainly on its pretrained knowledge.
- Important claims in the answer should be supported by the retrieved chunks.
- The augmented prompt should explicitly instruct the LLM to use the provided evidence.
- The model should avoid adding unsupported company policies, procedures, amounts, or timelines.
- Grounding reduces hallucination, but it does not guarantee that the answer is correct.
- The quality of the grounded answer depends strongly on the quality of the retrieved evidence.

Is the Retrieved Evidence Enough?

Criterion	Question
Relevance	Does the evidence relate to the user's question?
Coverage	Does it address all important parts of the question?
Support	Does it contain enough information to justify an answer?
Authority	Does it come from an approved and applicable source?
Currency	Is the information current and not superseded?
Consistency	Do the retrieved sources agree with each other?

Grounding: Answer Using the Retrieved Evidence

"I am attending a conference outside campus. Is there a ₱10,000 maximum registration fee that the University will reimburse?"

Relevant Documents



Chunk 1 — Trainings / Conferences
DV, approved ObR/BUR, invoice or statement of account, authority/approval to attend, Certificate of Participation/Attendance, and Narrative Report.
Chunk 2 — Outside-Campus Training
Additional requirements include a Certificate of Travel Completed and Itinerary of Travel.
Chunk 3 — General Reminder
The checklist contains minimum documentary requirements; other documents may be required when necessary.

LLM

"Yes. The University reimburses conference registration fees up to ₱10,000."

Poor Response

"The retrieved checklist identifies the supporting documents required for trainings and conferences, including the invoice or statement of account and approval to attend. However, it does not specify a ₱10,000 maximum reimbursable registration fee. That amount cannot be confirmed from the available evidence."

Better Controlled Answer

Classifying Evidence: Sufficient, Partial, or Insufficient

- Sufficient
  - All important parts of the question can be answered from reliable retrieved evidence.
- Partial
  - Some parts of the question are supported, but one or more important parts are missing.
- Insufficient
  - The retrieved evidence cannot support the main question.

What Should the System Do?

Question:
“What supporting documents are required for a training or conference held outside campus?”

Evidence:
✓ DV
✓ Approved ObR/BUR
✓ Invoice / Statement of Account
✓ Authority / Approval to attend
Status: SUFFICIENT

Question:
“What supporting documents are required for a conference outside campus, and is there a ₱10,000 maximum registration fee that can be reimbursed?”

Evidence:
✓ Supporting-document requirements
✓ Additional outside-campus travel documents
✗ ₱10,000 maximum reimbursement limit
Status: PARTIAL

Question:
“Will the University give me a ₱500 daily allowance just for attending an outside-campus conference?”

Evidence:
✓ Training/conference documentary requirements
✓ Outside-campus travel documents
✗ ₱500 daily allowance policy
Status: INSUFFICIENT

Evidence Status	System Response
Sufficient	Answer using the evidence and provide citations.
Partial	Answer only the supported parts and identify what is missing.
Insufficient	Avoid speculation and take another action.

Implementing Evals

- Build a small evaluation set
- Create representative questions with an **expected evidence status**

Prompt	Actual Truth
“What supporting documents are required for a training or conference held outside campus?”	Sufficient
“Is there a ₱10,000 maximum registration fee that can be reimbursed	Partial
“Will the University give me a ₱500 daily allowance?”	Insufficient



Predicted Truth
Partial
Partial
Insufficient

- Evidence sufficiency should be tested against a labeled benchmark—not judged only by whether the retrieved chunks “look relevant.”

Grounding and Evidence Sufficiency

- **Exercise 6: Is the Evidence Enough?**
- **Objectives**
  - Retrieve knowledge from the persistent **SQLite knowledge store**.
  - Combine **BM25 and SBERT** using Reciprocal Rank Fusion.
  - Expand retrieved results with neighboring chunks to improve evidence coverage.
  - Use Gemma as an **evidence evaluator**, not as the final answer generator.
  - Classify retrieved evidence as **SUFFICIENT, PARTIAL, or INSUFFICIENT**.
  - Identify which requested items are supported and which are missing.
  - Prevent the system from answering questions that are not supported by retrieved evidence.

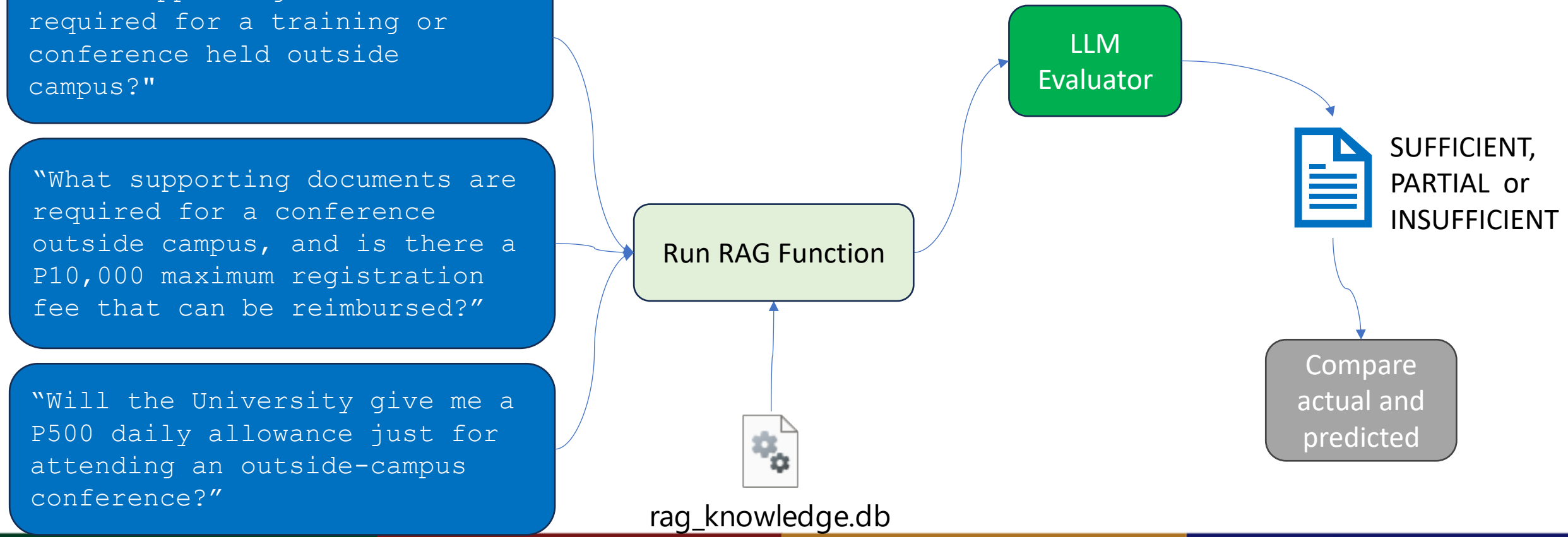
Grounding and Evidence Sufficiency

- **Exercise 6: Is the Evidence Enough?**
- **Business Case**

"What supporting documents are required for a training or conference held outside campus?"

"What supporting documents are required for a conference outside campus, and is there a P10,000 maximum registration fee that can be reimbursed?"

"Will the University give me a P500 daily allowance just for attending an outside-campus conference?"



Grounding and Evidence Sufficiency

```
#%% EXERCISE 6: GROUNDING AND EVIDENCE SUFFICIENCY
# Exercise 5 run_rag()
# → Retrieved Evidence
# → Gemma Evidence Evaluation
# → SUFFICIENT / PARTIAL / INSUFFICIENT
# → Compare with Expected Evaluation Label

from langchain_core.prompts import ChatPromptTemplate
from langchain_core.output_parsers import JsonOutputParser


if "run_rag" not in globals():

    raise RuntimeError(
        "run_rag() is not available.\n"
        "Run Exercise 5 first."
    )


if "llm" not in globals():

    raise RuntimeError(
        "The Gemma LLM is not available.\n"
        "Run Exercise 5 first."
    )


# ============================================================
# 1. EVIDENCE-EVALUATION PROMPT
# ============================================================

evidence_prompt = ChatPromptTemplate.from_messages(
    [
        (
            "system",
```

Grounding and Evidence Sufficiency

```
================================================================================
EVIDENCE EVALUATION TEST 3

USER QUESTION:
Will the University give me a P500 daily allowance just for attending an outside-campus conference?

EVIDENCE PROVIDED TO EVALUATOR:
[('C001', 1), ('C002', 1), ('C003', 1), ('C021', 7), ('C022', 8), ('C023', 8), ('C019', 7), ('C020', 7), ('C004', 2), ('C005', 2), ('C006', 2)]

EXPECTED STATUS:
INSUFFICIENT

PREDICTED STATUS:
INSUFFICIENT

SUPPORTED ITEMS:
['C001, p. 1: DV, Approved ObR/BUR, Purchase request, Travel Order approved by the Chancellor, Approval of funding/request to travel by HRDO
and Chancellor (as applicable), Invitation (Conferences, seminars, etc.), Promissory Note (settlement should be made 30 days from return date
for local travels, 60 days from return date for foreign travels)', 'C002, p. 1: “On or before _________________, I hereby promise to settle the
amount of P___________ in full. For failure to do so, I hereby authorize the University to deduct from my salary or any money due me. I also
acknowledge to have read COA Circular 96-004 dated April 19, 1996 of the Commission on Audit”,\n    ']

MISSING ITEMS:
[]

REASON:


EVALUATION RESULT:
PASS

================================================================================
EVIDENCE SUFFICIENCY EVALUATION SUMMARY

Tests Passed: 2/3
Classification Accuracy: 66.7%
```

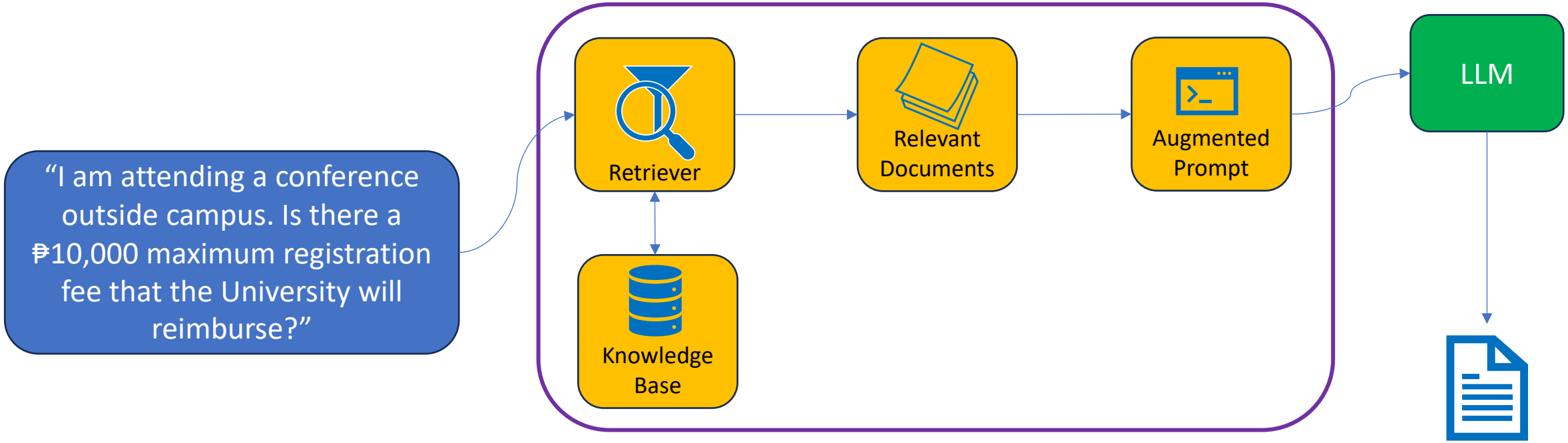
Outline for this Session

- Introduction to RAGs
- Preparing and Chunking Documents
- Keyword Search
- Semantic Similarity and Ranking
- Augmented Prompts
- Grounding and Evidence Sufficiency
- **Citations**
- Controlled RAGs
- Case Study and Project Work

Citation in RAG: Showing the Evidence

- **Definition:** A citation identifies the source used to support the generated answer.
- Citations help users **verify** where a statement came from.
- In RAG, citations usually originate from metadata attached to retrieved chunks.
- Useful citation metadata can include document title, section, page, date, version, and source ID.
- Citations are especially important in policy, compliance, technical support, healthcare, finance, and enterprise applications.
- A citation improves **transparency**, but it does not automatically make the answer correct.

Citations and Traceability

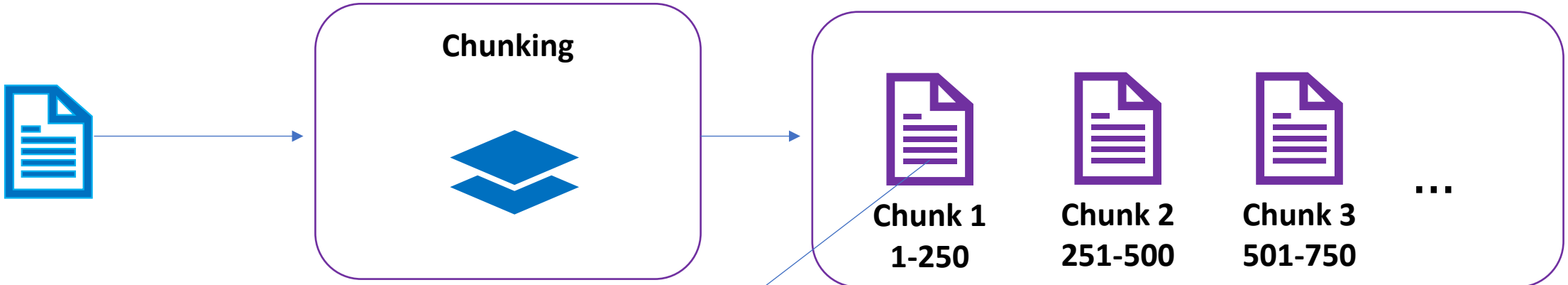


"The retrieved checklist specifies the supporting documents required for trainings and conferences, including the DV, approved ObR/BUR, invoice or statement of account, authority or approval to attend, Certificate of Participation/Attendance, and Narrative Report. For trainings outside campus, a Certificate of Travel Completed and Itinerary of Travel are additionally required. [1]
However, the retrieved evidence does **not specify a ₱10,000 maximum reimbursable registration fee**, so that amount cannot be confirmed from the available checklist. [1]"

Sources:

[1] Pre-Audit Checklist for Common Transactions, Trainings/Conferences, p. 8

How Citation Information Is Preserved




```
{
  "text": "...",
  "document": “Reimbursement",
  "section": "3.2",
  "page": 7
}
```

Citation Evaluation

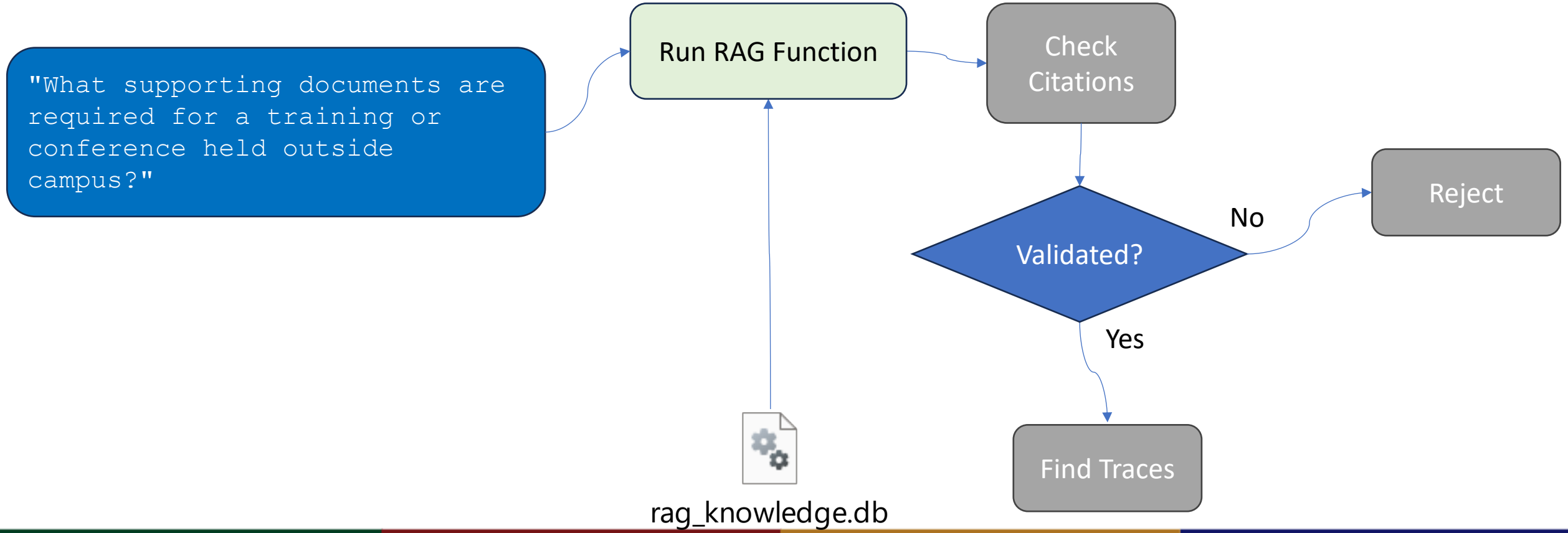
Criterion	Question
Presence	Is a source cited for the important claim?
Correctness	Does the cited source actually support the claim?
Completeness	Are all important factual claims adequately cited?
Authority	Is the source approved and appropriate for the question?
Currency	Is the source current and not superseded?
Traceability	Can the user locate the original document or section?

Citations and Traceability

- **Exercise 7: Cite and Verify Retrieved Sources**
- **Objectives**
  - Preserve **chunk ID, page number, and source metadata**.
  - Generate a grounded answer using **Gemma**.
  - Require citations for factual documentary requirements.
  - Verify that citations refer only to evidence actually retrieved.
  - Trace a generated claim back to the original PDF.

Citations and Traceability

- **Exercise 7: Cite and Verify Retrieved Sources**
- **Business Case**



Exercise 7 — Citations

```python
#%% EXERCISE 7: CITATIONS AND TRACEABILITY
# Exercise 5 run_rag()
# → Citation-Aware RAG Answer
# → Citation Validation
# → Traceability
import re
if "run_rag" not in globals():

    raise RuntimeError(
        "run_rag() is not available.\n"
        "Run Exercise 5 first."
    )


# ============================================================
# HELPER FUNCTION: VALIDATE CITATIONS
# ============================================================

def validate_citations(
    answer,
    retrieved_docs
):

    # ------------------------------------------------------------
    # 1. EXTRACT CITATIONS FROM ANSWER
    # ------------------------------------------------------------
    # Expected format:
    # [C023, p. 8]

    citation_matches = re.findall(
        r"\[(C\d+),\s*p\.\s*(\d+)\]",
        answer
    )

    cited_citations = set(
        citation_matches
    )


    # ------------------------------------------------------------
    # 2. BUILD SET OF RETRIEVED CITATIONS
    # ------------------------------------------------------------

    retrieved_citations = set(
        (
            doc.metadata["chunk_id"],
            str(doc.metadata["page_number"])
        )
```

```
CHUNKS AFTER NEIGHBOR EXPANSION:
['C022', 'C023', 'C024', 'C018', 'C019', 'C020', 'C021', 'C004', 'C005', 'C006']

================================================================================
RAG RESPONSE:
--------------------------------------------------------------------------------
The following supporting documents are required for Trainings outside of campus: [C022, p. 8]

*   Certificate of travel completed [C023, p. 8]
*   Itinerary of travel [C023, p. 8]


================================================================================
CITATION CHECK:
PASSED

REASON:
All cited chunk IDs and page numbers appear in the retrieved evidence.

CITATIONS FOUND:
[C022, p. 8]
[C023, p. 8]

INVALID CITATIONS:
None

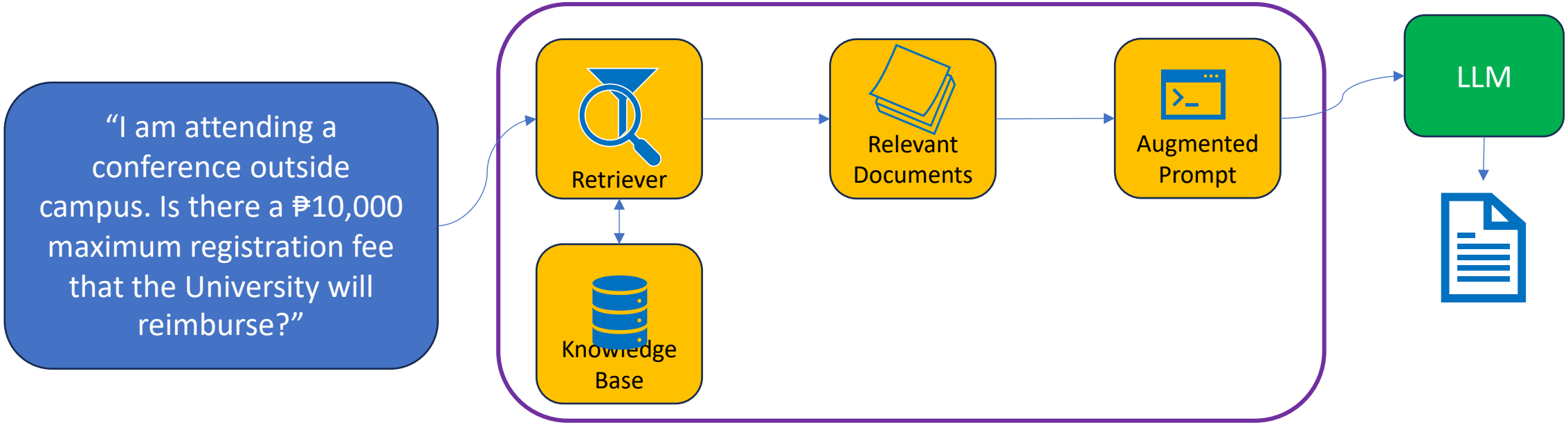
================================================================================
TRACEABILITY:
C022 | Page: 8 | Source: PreAudit Checklist for Common Transactions.pdf | Cited: True
C023 | Page: 8 | Source: PreAudit Checklist for Common Transactions.pdf | Cited: True
C024 | Page: 9 | Source: PreAudit Checklist for Common Transactions.pdf | Cited: False
C018 | Page: 6 | Source: PreAudit Checklist for Common Transactions.pdf | Cited: False
C019 | Page: 7 | Source: PreAudit Checklist for Common Transactions.pdf | Cited: False
C020 | Page: 7 | Source: PreAudit Checklist for Common Transactions.pdf | Cited: False
C021 | Page: 7 | Source: PreAudit Checklist for Common Transactions.pdf | Cited: False
C004 | Page: 2 | Source: PreAudit Checklist for Common Transactions.pdf | Cited: False
C005 | Page: 2 | Source: PreAudit Checklist for Common Transactions.pdf | Cited: False
C006 | Page: 2 | Source: PreAudit Checklist for Common Transactions.pdf | Cited: False
```

Outline for this Session

- Introduction to RAGs
- Preparing and Chunking Documents
- Keyword Search
- Semantic Similarity and Ranking
- Augmented Prompts
- Grounding and Evidence Sufficiency
- Citations
- **Controlled RAGs**
- Case Study and Project Work

Basic versus Controlled RAG

- **Basic RAG:** Retrieval is performed; relevant chunks are added to the prompt and the LLM generates an answer.
  - Few controls may exist around unsupported claims.
  - Missing evidence may still result in a generated answer.
  - The output may not be validated.
  - Citations may not be included.



A Basic RAG Needs More Control

- A RAG pipeline can retrieve information and still **fail** in several ways:
  - The retrieved chunks are related to the question but do not actually answer it.
  - Only part of a multi-part question is supported.
  - The system retrieves an outdated policy.
  - Two retrieved documents contain conflicting information.
  - A citation is present, but it does not support the generated claim.
  - The LLM adds information that was not contained in the retrieved evidence.
  - The question requires customer-specific or real-time information that the knowledge base cannot provide.

Basic versus Controlled RAG

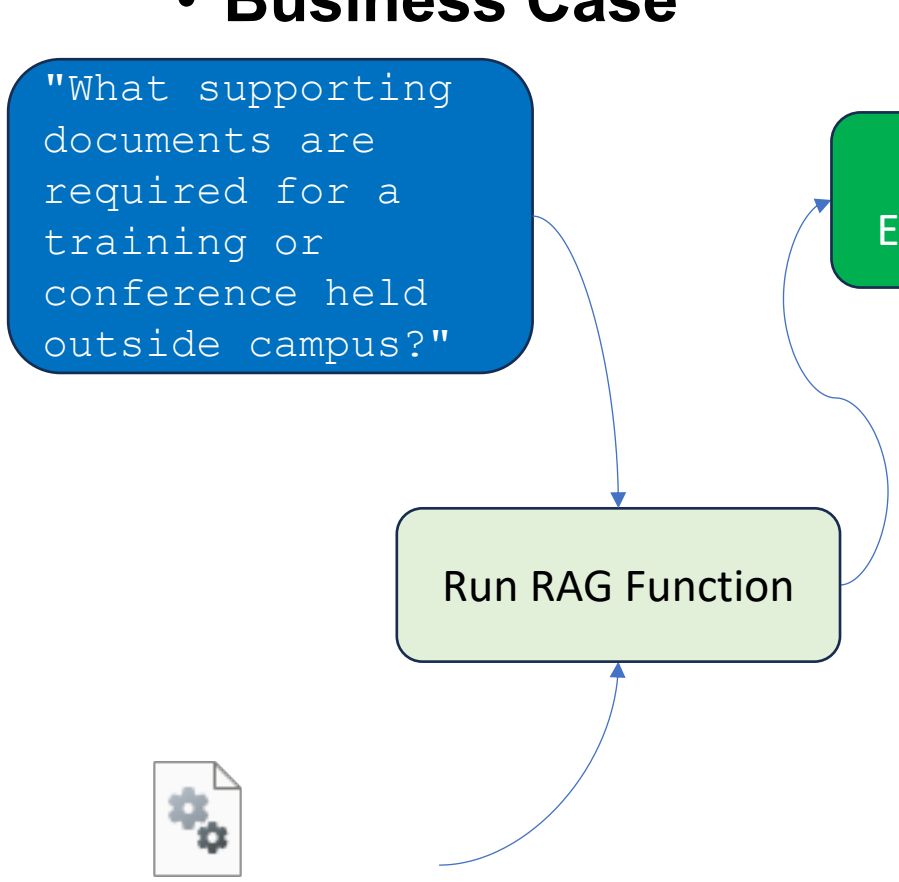
- **Controlled RAG** adds checks that determine **whether and how the system should answer**.
- Controlled RAG Includes:
  - Relevance thresholds.
  - Evidence sufficiency checks.
  - Grounding instructions.
  - Citation requirements.
  - Structured output.
  - Pydantic validation.
  - Business rules.
  - Human escalation.
  - Logging and traceability.

Controlled RAG

- **Exercise 8: From Basic RAG to Controlled RAG**
- **Objectives**
  - Apply explicit rules for releasing AI-generated answers.
  - Abstain when retrieved evidence is insufficient.
  - Require human review for partial or failed responses.
  - Evaluate both evidence classification and control decisions.

Citations and Traceability

- **Exercise 8: From Basic RAG to Controlled RAG**
- **Business Case**



Evidence	Citation	Decision
SUFFICIENT	PASSED	RELEASE
SUFFICIENT	FAILED	DO NOT RELEASE
PARTIAL	PASSED	HUMAN REVIEW BEFORE RELEASE
PARTIAL	FAILED	DO NOT RELEASE
INSUFFICIENT	—	ABSTAIN

```python
#%% EXERCISE 8: CONTROLLED RAG
# Exercise 5 run_rag()
# → Exercise 6 Evidence Sufficiency
# → Exercise 7 Citation Validation
# → Release / Abstain / Human Review


# ============================================================
# REQUIREMENTS
# ============================================================
# Run Exercises 5, 6, and 7 first.
#
# Exercise 5 provides:
#     run_rag()
#
# Exercise 6 provides:
#     evidence_chain
#
# Exercise 7 provides:
#     validate_citations()
#
# Exercise 8 adds:
#     CONTROL DECISIONS


if "run_rag" not in globals():

    raise RuntimeError(
        "run_rag() is not available.\n"
        "Run Exercise 5 first."
    )


if "evidence_chain" not in globals():

    raise RuntimeError(
        "evidence_chain is not available.\n"
        "Run Exercise 6 first."
    )


if "validate_citations" not in globals():

    raise RuntimeError(
        "validate_citations() is not available.\n"
        "Run Exercise 7 first."
    )
```

```
================================================================================
CONTROL DECISION:
ABSTAIN

RAG DRAFT RELEASED:
NO

FINAL RESPONSE:
The retrieved checklist evidence does not contain enough information to answer this question.

HUMAN REVIEW REQUIRED:
YES

EXPECTED STATUS:
INSUFFICIENT

PREDICTED STATUS:
INSUFFICIENT

STATUS TEST:
PASS

EXPECTED CONTROL DECISION:
ABSTAIN

PREDICTED CONTROL DECISION:
ABSTAIN

CONTROL TEST:
PASS

================================================================================
CONTROLLED RAG TEST SUMMARY

Evidence Status Tests Passed: 2/3
Evidence Classification Accuracy: 66.7%

Control Decision Tests Passed: 2/3
Control Decision Accuracy: 66.7%
```

Outline for this Session

- Introduction to RAGs
- Preparing and Chunking Documents
- Keyword Search
- Semantic Similarity and Ranking
- Augmented Prompts
- Grounding and Evidence Sufficiency
- Citations
- Controlled RAGs
- **Case Study and Project Work**

Case Study and Project Work

- **Objective:** Develop a controlled RAG application that retrieves academic policies from the UP Diliman handbook, generates evidence-grounded answers, and validates citations before releasing responses.
- Tasks:
  - Load the academic handbook, split it into chunks, and store page-aware information in SQLite.
  - Implement hybrid retrieval using BM25, SBERT, and Reciprocal Rank Fusion (RRF).
  - Expand retrieved evidence by including neighboring chunks.
  - Create a reusable run_rag(question) function using Ollama and Gemma 3.
  - Evaluate evidence sufficiency and verify source citations.
  - Apply Python controls to determine whether to RELEASE, REVIEW, or ABSTAIN.

Case Study 4: Page-Aware Academic Policy RAG

End-to-End Workflow: From Knowledge Base Construction to Controlled Response

Overview

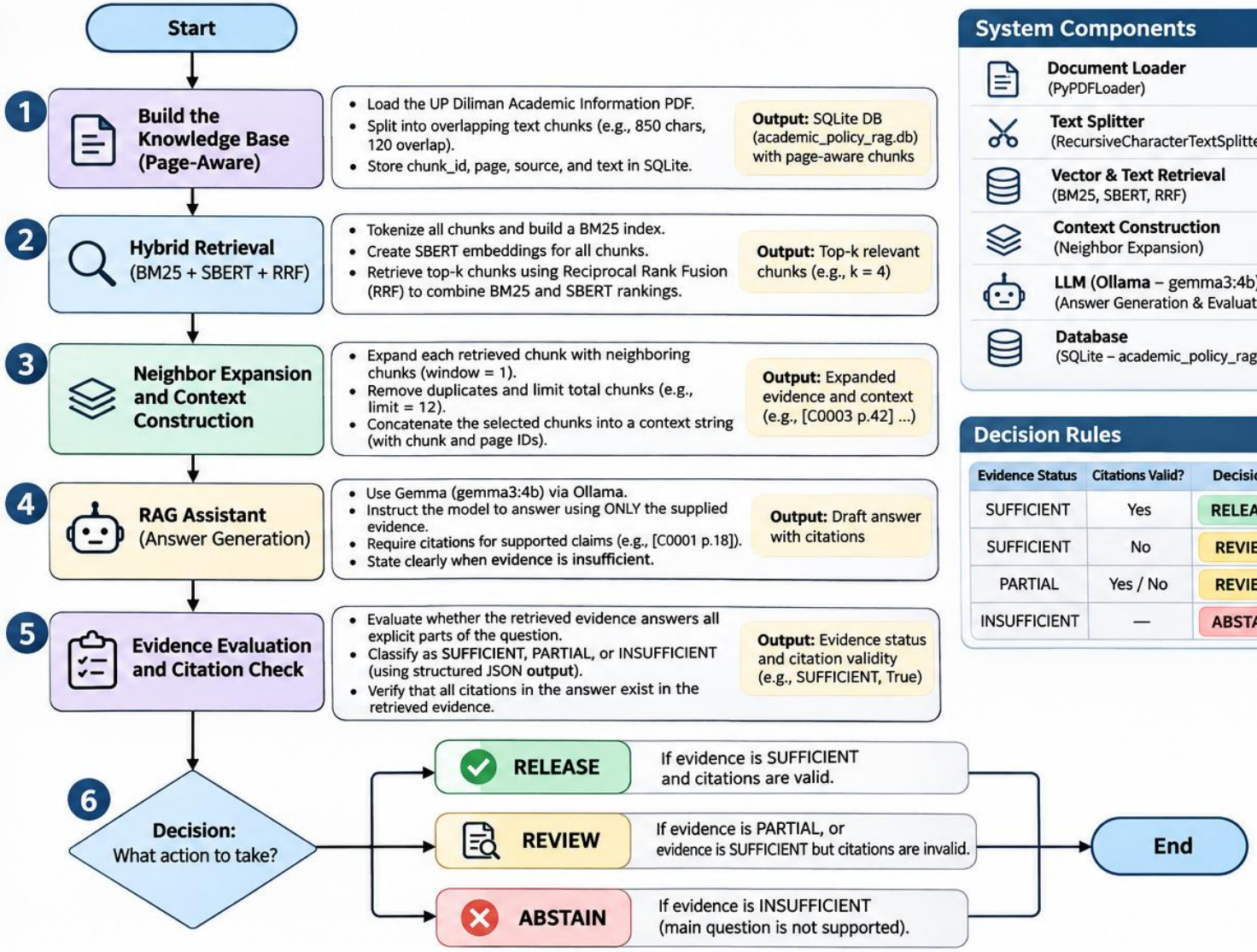
Build a controlled Retrieval-Augmented Generation (RAG) system that answers student questions about UP Diliman academic policies using the Academic Information handbook.

Workflow Steps (Cells 1–6)

1. Build the knowledge base (page-aware chunks)
2. Retrieve relevant chunks (BM25 + SBERT + RRF)
3. Expand with neighboring chunks and build context
4. Generate an answer (RAG Assistant)
5. Evaluate evidence sufficiency and check citations
6. Make a controlled decision (Release / Review / Abstain)

Key Outputs

- ✓ SQLite knowledge base (academic_policy_rag.db)
- ✓ Retrieved and expanded evidence (with chunk/page IDs)
- ✓ Grounded answer with citations (e.g., [C0003 p.42])
- ✓ Evidence status (SUFFICIENT / PARTIAL / INSUFFICIENT)
- ✓ Final decision (RELEASE / REVIEW / ABSTAIN)



System Components

- **Document Loader** (PyPDFLoader)
- **Text Splitter** (RecursiveCharacterTextSplitter)
- **Vector & Text Retrieval** (BM25, SBERT, RRF)
- **Context Construction** (Neighbor Expansion)
- **LLM (Ollama** – gemma3:4b) (Answer Generation & Evaluation)
- **Database** (SQLite – academic_policy_rag.db)

Decision Rules

Evidence Status	Citations Valid?	Decision
SUFFICIENT	Yes	RELEASE
SUFFICIENT	No	REVIEW
PARTIAL	Yes / No	REVIEW
INSUFFICIENT	—	ABSTAIN

Capstone Project Case: Integrate RAG into Your AI Project

- Objective: Extend the controlled LLM component from Milestone 3 by integrating knowledge retrieval from documents or other approved information sources.
- Group Instructions:
  - Select an activity in the Milestone 2 swimlane flowchart that requires organizational knowledge or reference documents.
  - Identify an authoritative document or knowledge source relevant to the selected activity.
  - Adapt the guided case code to prepare, chunk, and retrieve information from your own source.
  - Integrate retrieval and prompt augmentation into the existing Milestone 3 LLM function.
  - Implement evidence sufficiency and source citation controls.
  - Test the application using answerable and unanswerable business questions.

Outline for this Session

- Introduction to RAGs
- Preparing and Chunking Documents
- Keyword Search
- Semantic Similarity and Ranking
- Augmented Prompts
- Grounding and Evidence Sufficiency
- Citations
- Controlled RAGs
- Case Study and Project Work